

Instalasi Flutter

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Outlines

1. Pengantar instalasi flutter
2. Daftar file pendukung instalasi flutter
3. Instalasi Java Development Kit
4. Instalasi Flutter SDK
5. Instalasi VSCode
6. Instalasi Extensions VSCode (Dart dan Flutter)
7. Instalasi Android SDK
8. Instalasi physical device (wired)
9. Instalasi physical device (wireless)
10. Instalasi virtual device (emulator)

Capaian Pembelajaran

Setelah materi ini disampaikan, mahasiswa diharapkan mampu

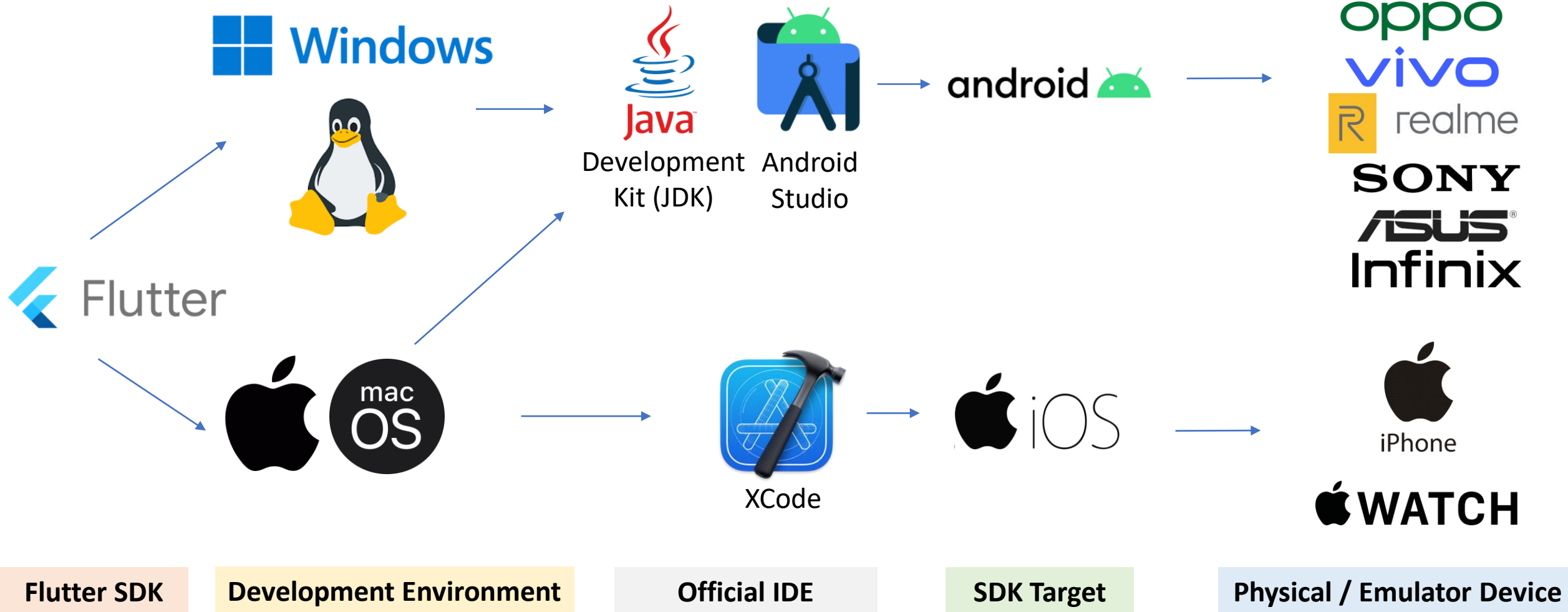
- a. Menjelaskan lingkungan pengembangan Aplikasi flutter
- b. Menjelaskan file-file yang dibutuhkan untuk instalasi lingkungan pengembangan Aplikasi flutter
- c. Melakukan instalasi lingkungan pengembangan Aplikasi flutter
- d. Menjalankan Aplikasi pada physical device maupun virtual device

1. Pengantar instalasi flutter

1. Pengantar instalasi flutter

- Komponen utama dalam mengembangkan Aplikasi mobile flutter adalah **Flutter** Software Development Kit (**SDK**)
 - SDK adalah seperangkat tools (*software suite*) lengkap yang mencakup semua kebutuhan dalam mengembangkan suatu Aplikasi
- Mengingat flutter adalah framework Aplikasi mobile **cross-platform native**, maka dibutuhkan **SDK target** (tujuan akhir) suatu sistem operasi mobile, misalnya Android atau iOS
 - SDK target (Android atau iOS) masing-masing bisa diperoleh melalui instalasi IDE resmi, yaitu **Android Studio** dan **XCode**

1. Pengantar instalasi flutter



1. Pengantar instalasi flutter

- **SDK Target** dapat diperoleh melalui **instalasi IDE**
- **Kendala umum** pada pengembangan yang seringkali ditemui:

iOS

- a. Emulator iOS hanya dapat berjalan di macOS
- b. Physical device (iphone) harganya relatif mahal
- c. Terbatasnya akses pada perangkat macOS (imac, macbook)
- d. XCode IDE hanya dapat digunakan pada macOS versi terbaru (dibutuhkan storage besar)

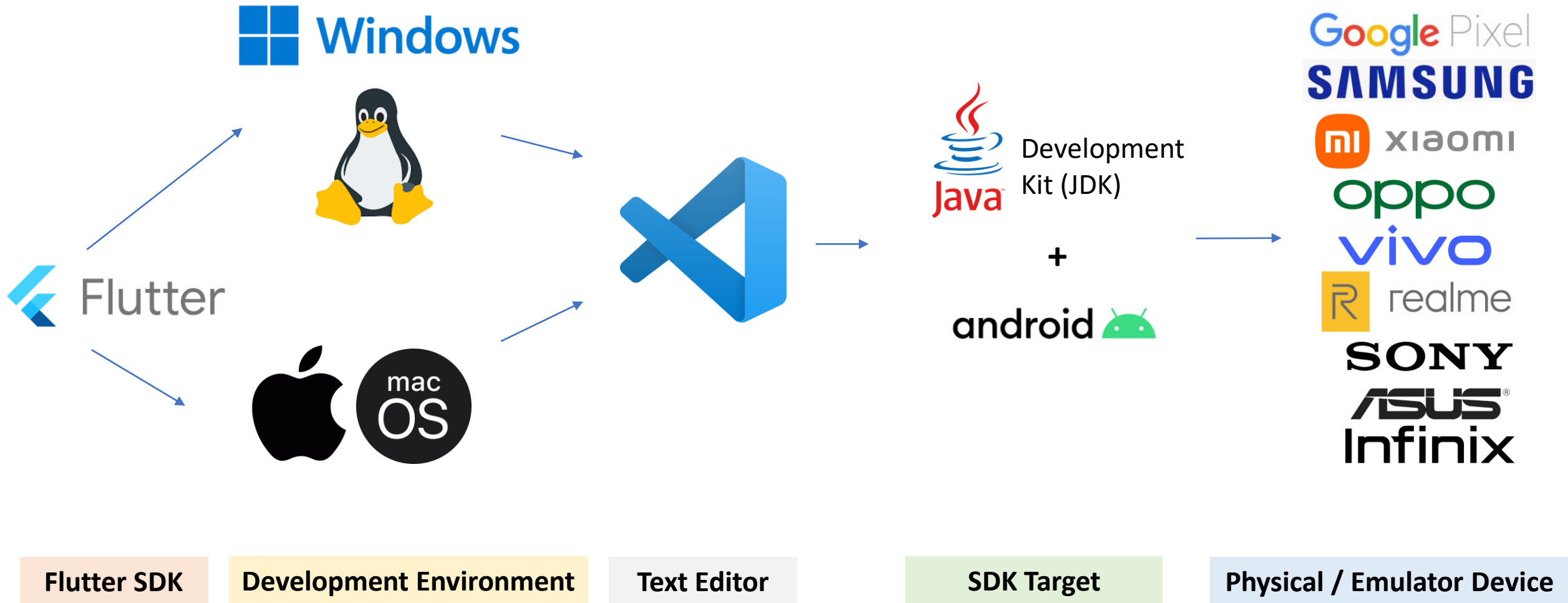
Android

- a. Dibutuhkan kapasitas storage dan RAM besar untuk menggunakan Android Studio IDE secara ideal (lancar)
- b. Emulator Android menggunakan kapasitas RAM besar

1. Pengantar instalasi flutter

- Pembelajaran **berfokus** pada pengembangan Aplikasi mobile dengan sistem operasi **Android**
- Disarankan untuk menggunakan **physical device** (smartphone masing-masing)
- Jika tidak ada physical device, gunakan **emulator device** (diasumsikan perangkat telah memenuhi prasyarat)
- **VSCode** digunakan sebagai alternatif yang lebih ringan dari Android Studio IDE

1. Pengantar instalasi flutter



2. Daftar file pendukung instalasi flutter

2. Daftar file pendukung instalasi flutter

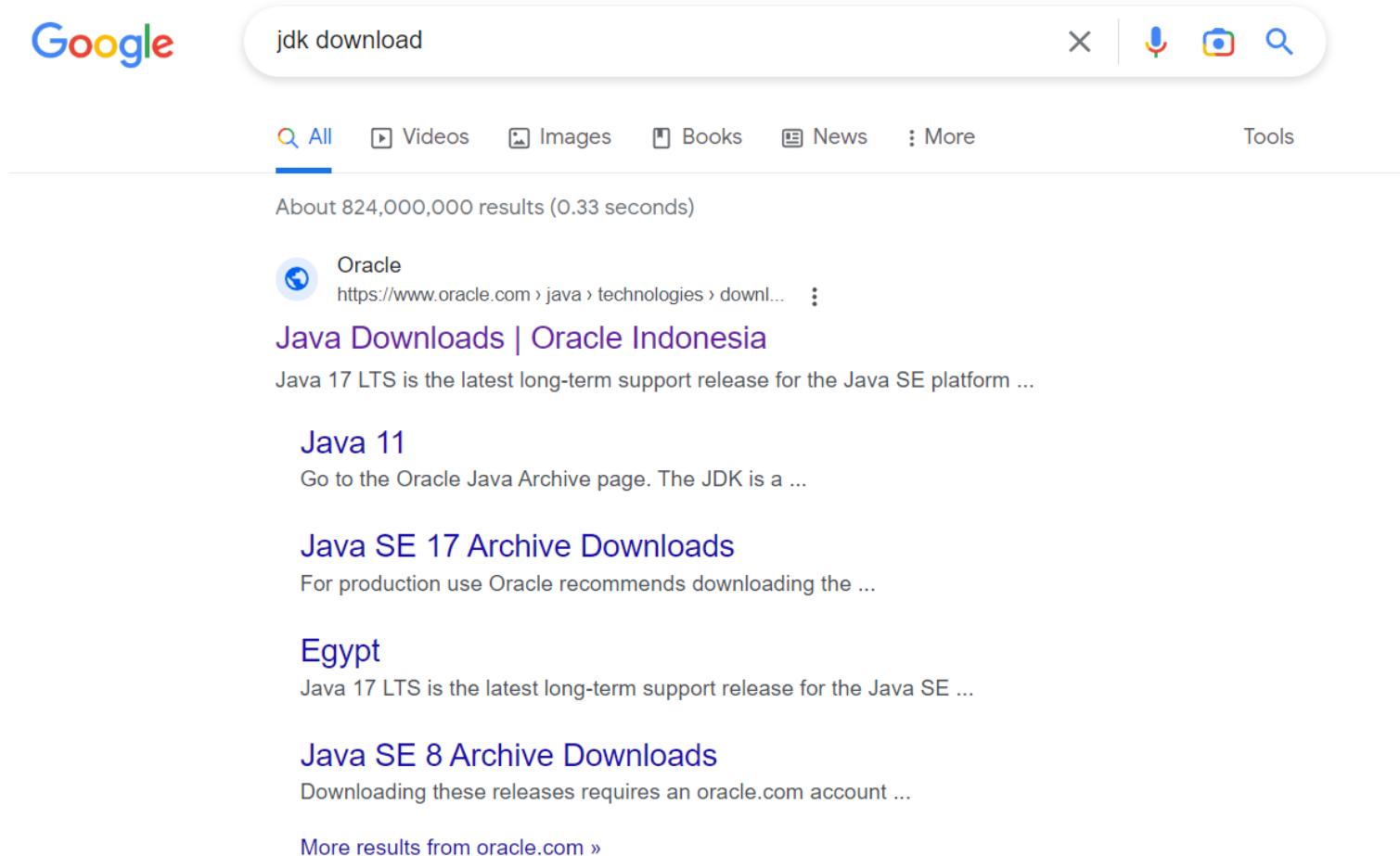
- a. Java Development Kit (JDK)
- b. Flutter SDK
- c. VSCode
 1. Extension Dart
 2. Extension Flutter
- d. Android SDK
 1. command line tools
 - platform-tools
 - build-tools
 - SDK-tools
 - Android Debug Bridge (ADB)
 - Android emulator

#15Feb2023

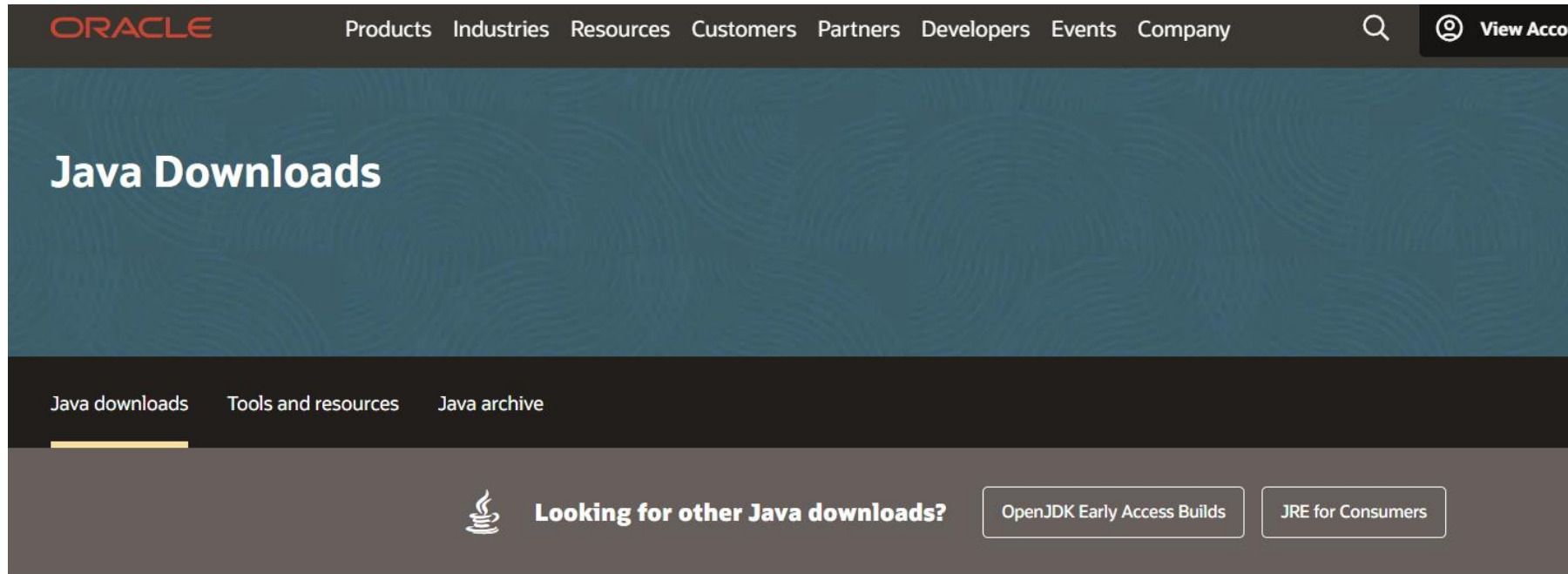
Name ^	
	commandlinetools-win-9477386_latest.zip
	flutter_windows_3.7.3-stable.zip
	jdk-17_windows-x64_bin.exe
	VSCodeUserSetup-x64-1.75.0.exe

3. Instalasi Java Development Kit

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3. Instalasi Java Development Kit



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1

Java 17 LTS is the latest long-term support release for the Java SE platform. JDK 19 and JDK 17 binaries are free to use in production and free to redistribute, at no cost, under the [Oracle No-Fee Terms and Conditions](#).

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Java 19

Java 17

2

Java SE Development Kit 19.0.2 downloads

3. Instalasi Java Development Kit

[Java downloads](#) [Tools and resources](#) [Java archive](#)

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Java SE Development Kit 17.0.6 downloads

Thank you for downloading this release of the Java™ Platform, Standard Edition Development Kit (JDK™). The JDK is a development environment for building applications and programming language.

The JDK includes tools for developing and testing programs written in the Java programming language and running on the Java platform.

[Linux](#) [macOS](#) [Windows](#)

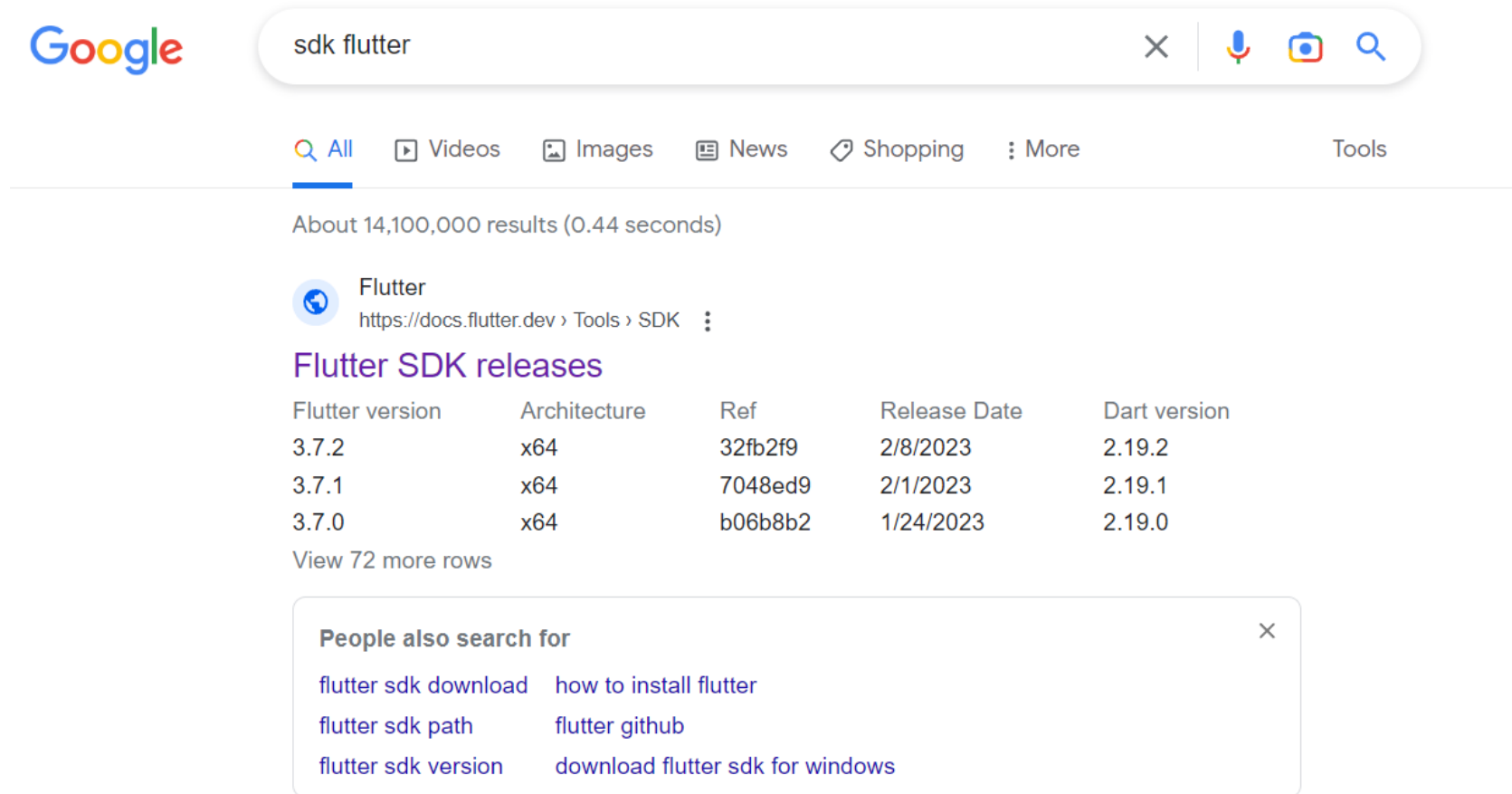
Product/file description	File size	Download
x64 Compressed Archive	172.11 MB	https://download.oracle.com/java/17/latest/jdk-17_windows-x64_bin.zip (sha256)
x64 Installer	153.22 MB	https://download.oracle.com/java/17/latest/jdk-17_windows-x64_bin.exe (sha256)
x64 MSI Installer	152.01 MB	https://download.oracle.com/java/17/latest/jdk-17_windows-x64_bin.msi (sha256)

3. Instalasi Java Development Kit

- Download JDK
- Install
- Lakukan set environment variable dengan path:
C:\Program Files\Java\jdk-17.x.x\bin

4. Instalasi Flutter SDK

4. Instalasi Flutter SDK



Google search results for "sdk flutter". The search bar shows "sdk flutter" with a clear button (X) and icons for voice search, image search, and a magnifying glass. Below the search bar are tabs for "All", "Videos", "Images", "News", "Shopping", "More", and "Tools". The "All" tab is selected.

About 14,100,000 results (0.44 seconds)

Flutter
<https://docs.flutter.dev> > Tools > SDK

Flutter SDK releases

Flutter version	Architecture	Ref	Release Date	Dart version
3.7.2	x64	32fb2f9	2/8/2023	2.19.2
3.7.1	x64	7048ed9	2/1/2023	2.19.1
3.7.0	x64	b06b8b2	1/24/2023	2.19.0

[View 72 more rows](#)

People also search for

- flutter sdk download
- how to install flutter
- flutter sdk path
- flutter github
- flutter sdk version
- download flutter sdk for windows

4. Instalasi Flutter SDK

The screenshot shows the Flutter SDK releases page. A red box labeled '1' highlights the 'Windows' tab in the platform selection area. Another red box labeled '2' highlights the '3.7.3' version in the table of releases.

Flutter

Multi-Platform ▾ Development ▾ Ecosystem ▾ Showcase Docs ▾ 🔍 🐦

Flutter SDK releases

[Tools](#) > [SDK](#) > Releases

The Stable channel contains the most stable Flutter builds. See [Flutter's channels](#) for details.

1 Windows macOS Linux

Stable channel (Windows)

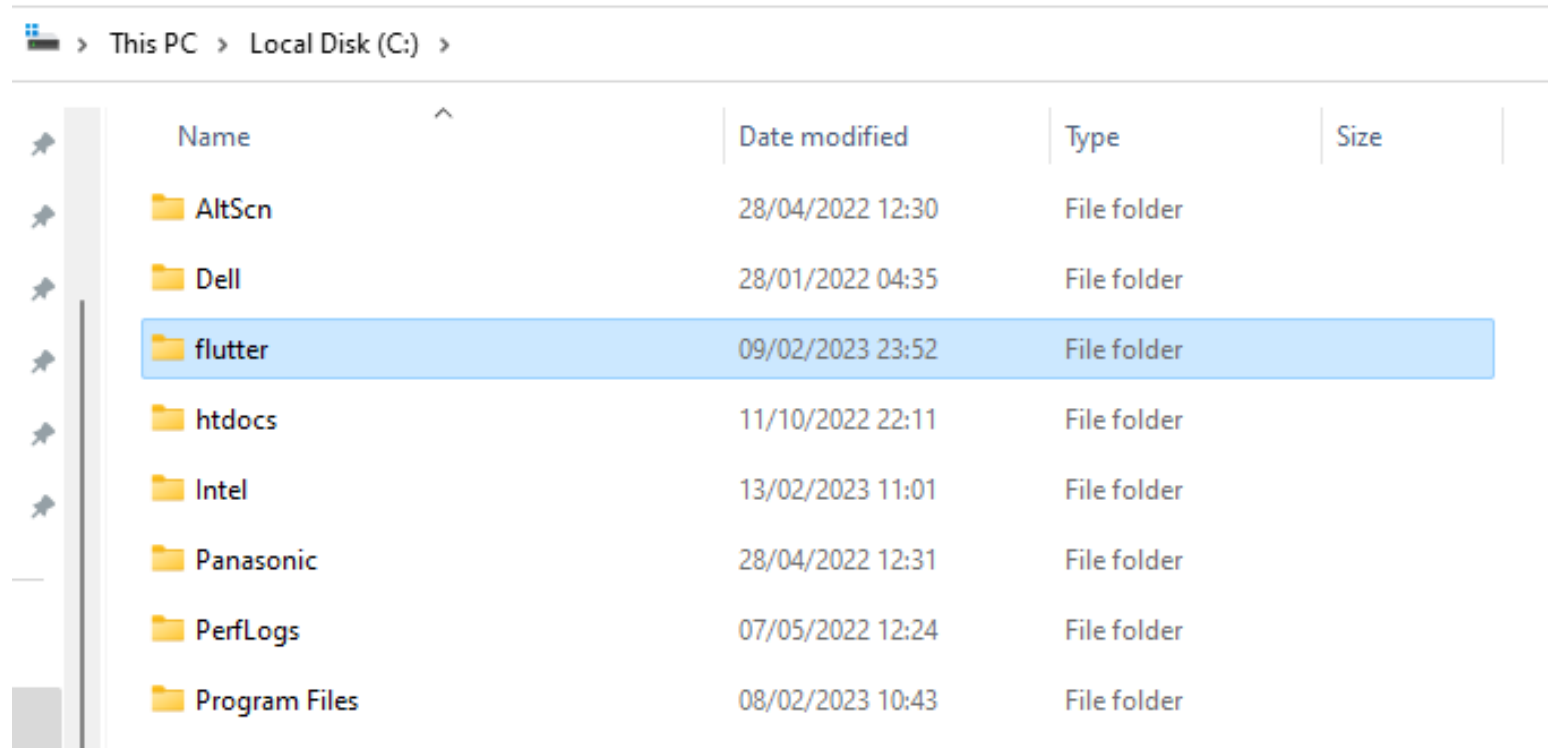
Select from the following scrollable list:

Flutter version	Architecture	Ref	Release Date	Dart version
3.7.3	x64	9944297	2/10/2023	2.19.2
3.7.2	x64	32fb2f9	2/9/2023	2.19.2
3.7.1	x64	7048ed9	2/2/2023	2.19.1

Get started ▾
Samples & tutorials ▾
Development ▴
 ▶ User interface
 ▶ Data & backend
 ▶ Accessibility & internationalization
 ▶ Platform integration
 ▶ Packages & plugins
 ▶ Add Flutter to an existing app
 ▼ Tools & features
 Android Studio & IntelliJ
 Visual Studio Code
 ▶ DevTools
 ▼ Flutter SDK
 Overview
 Upgrading
 [Releases](#)
 Breaking changes

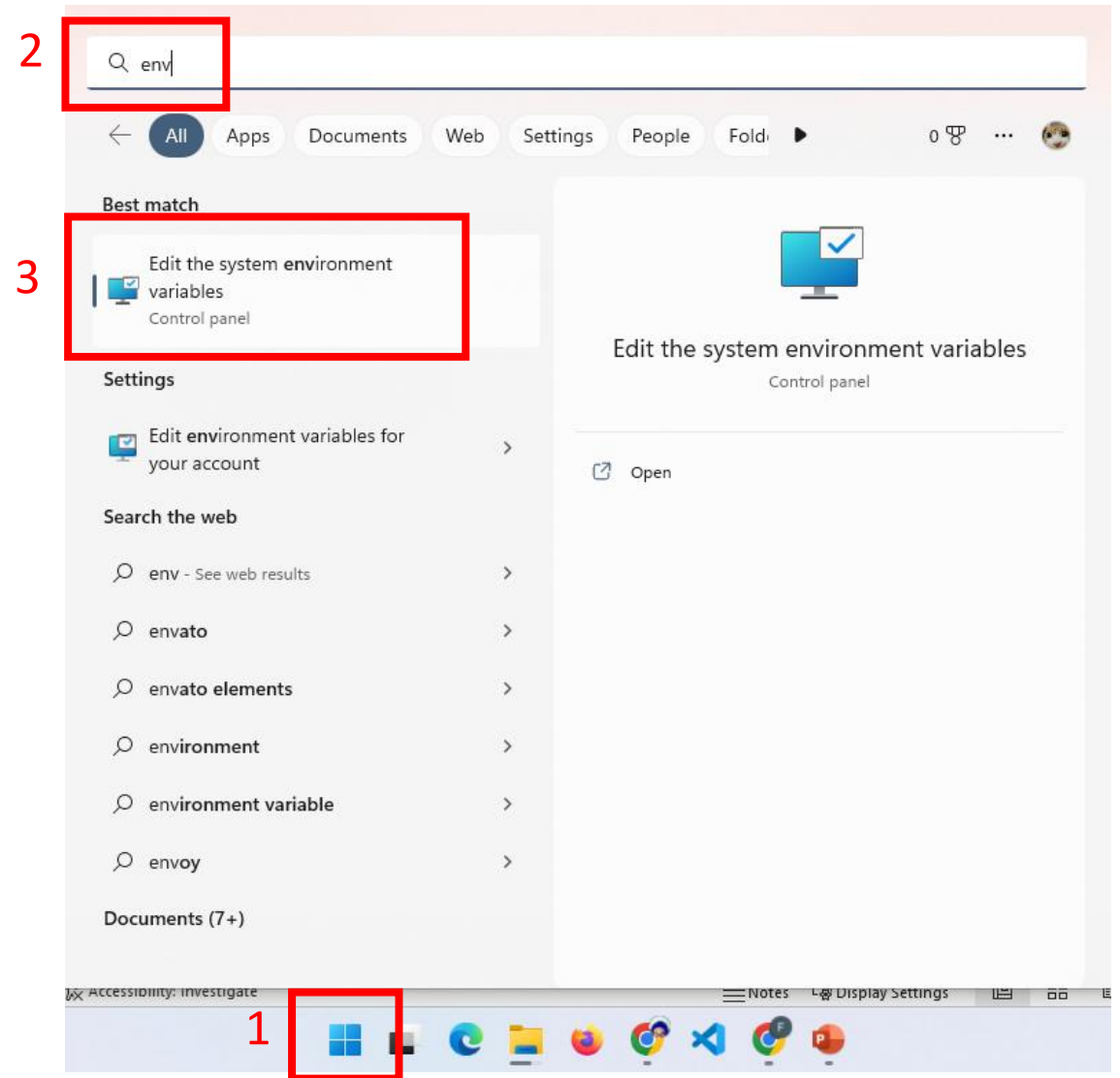
4. Instalasi Flutter SDK

- Extract file *.zip ke Drive C:



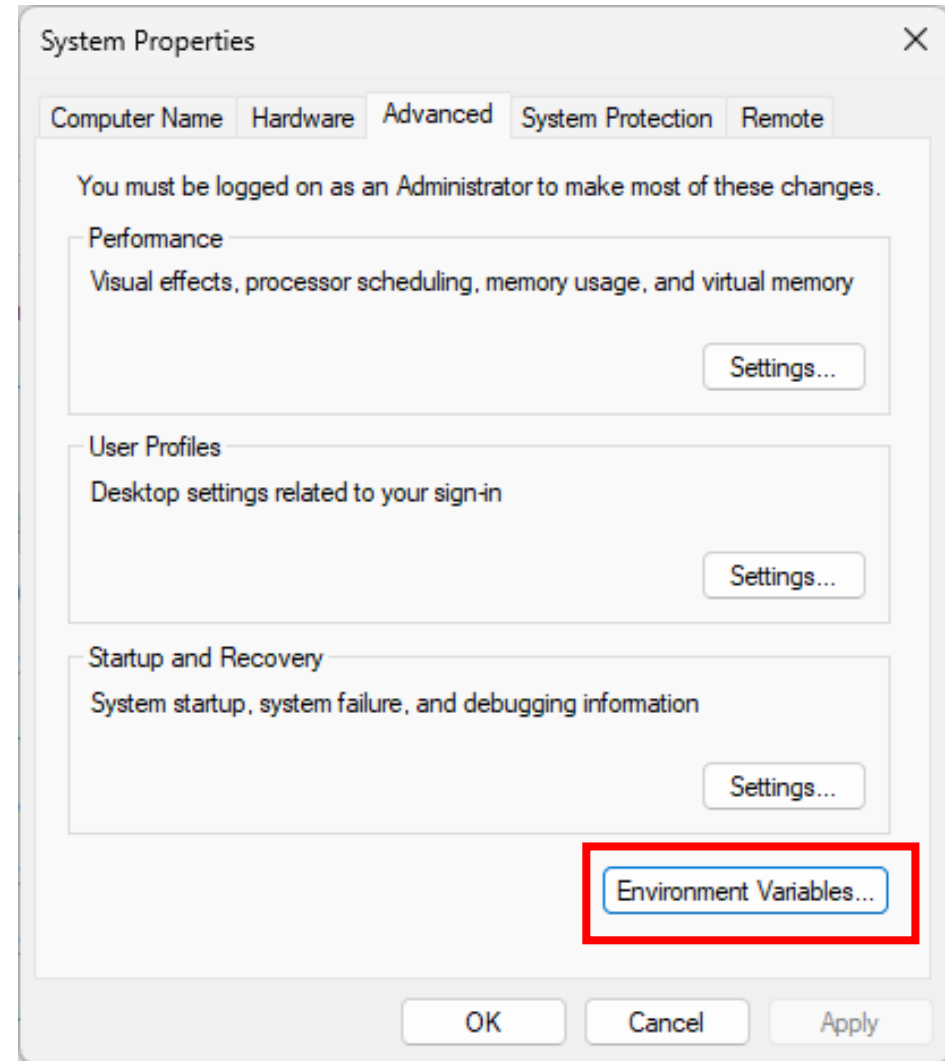
4. Instalasi Flutter SDK

- Set environment variable dengan path lokasi C:\flutter\bin

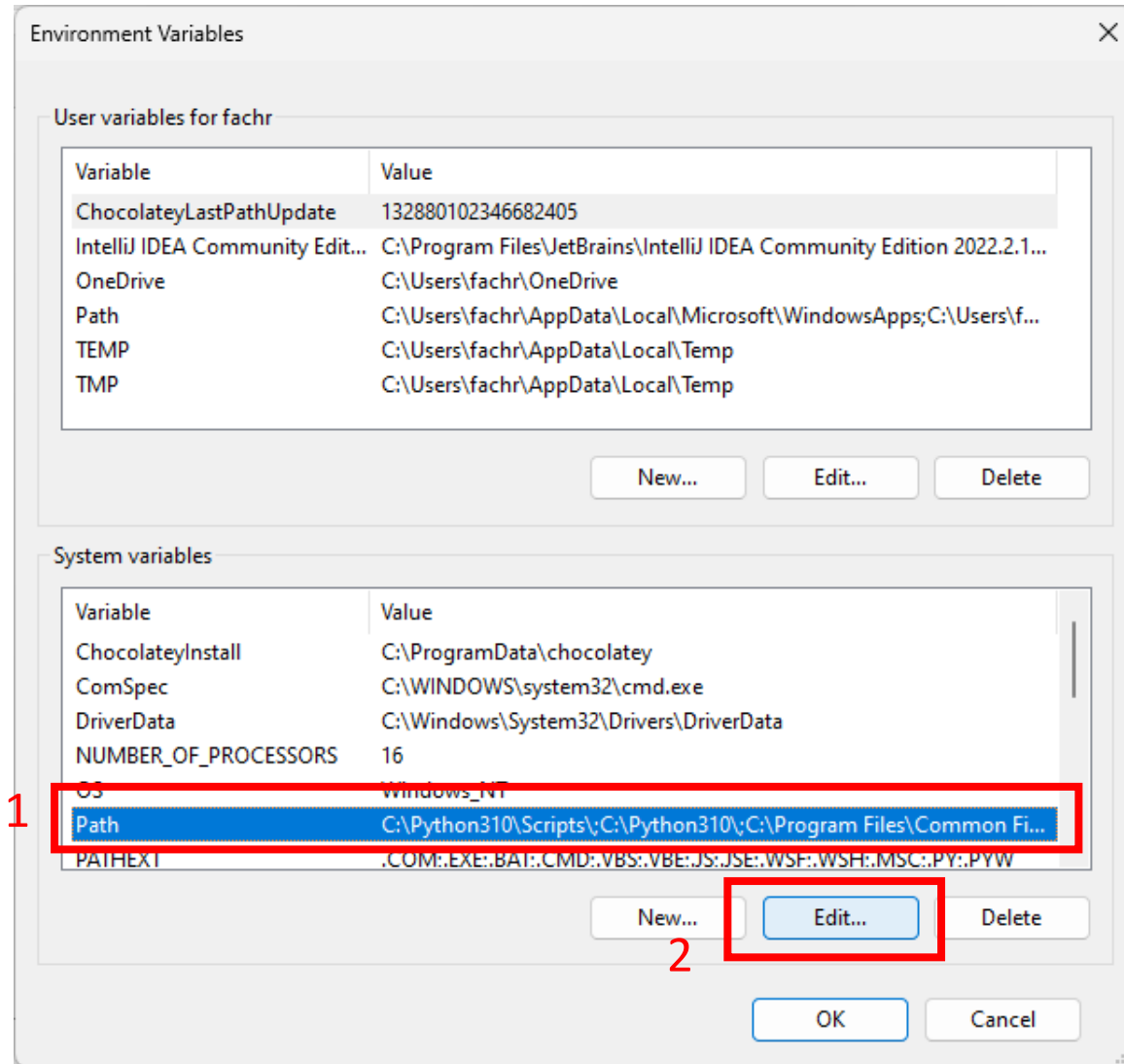


4. Instalasi Flutter SDK

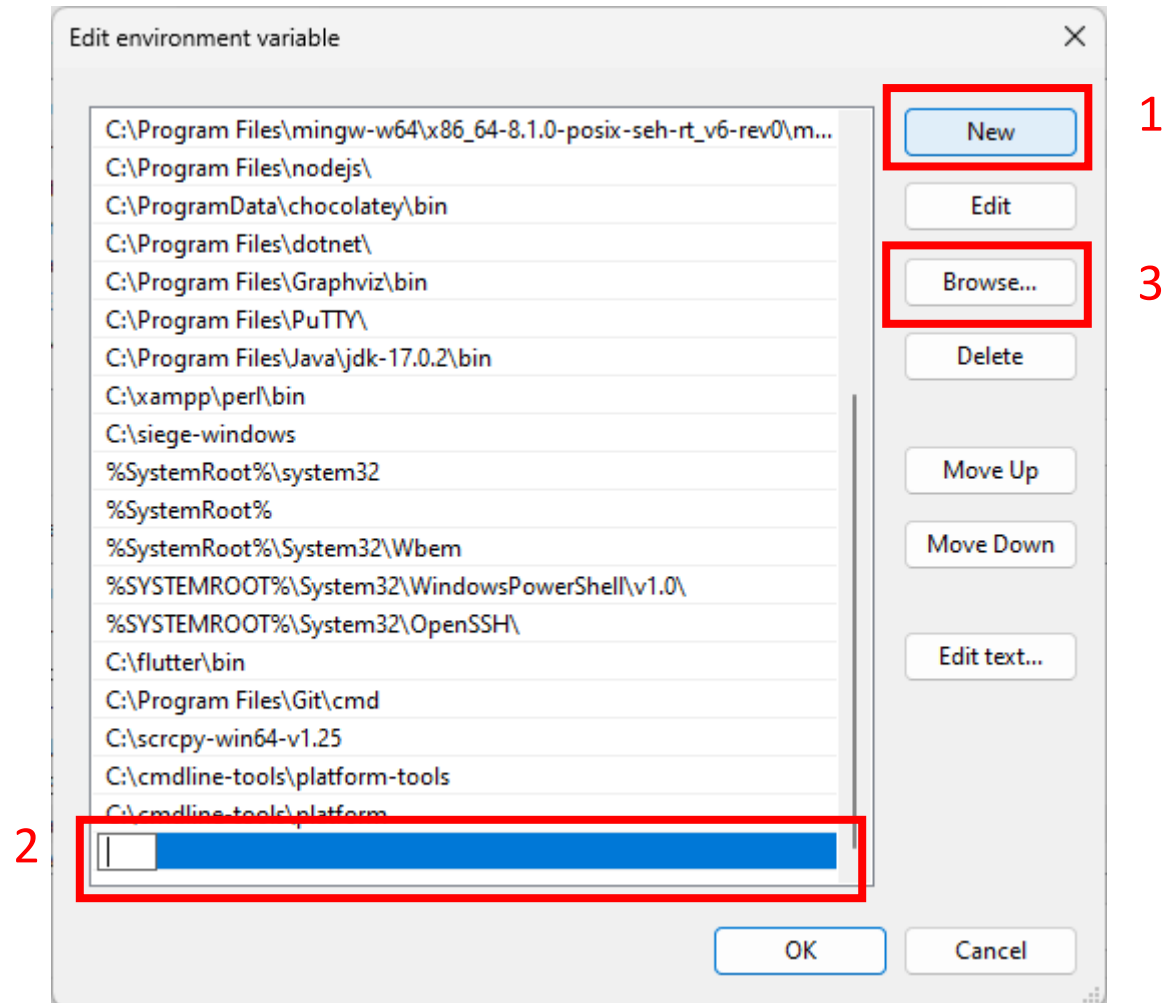
- Set environment variable dengan path lokasi C:\flutter\bin



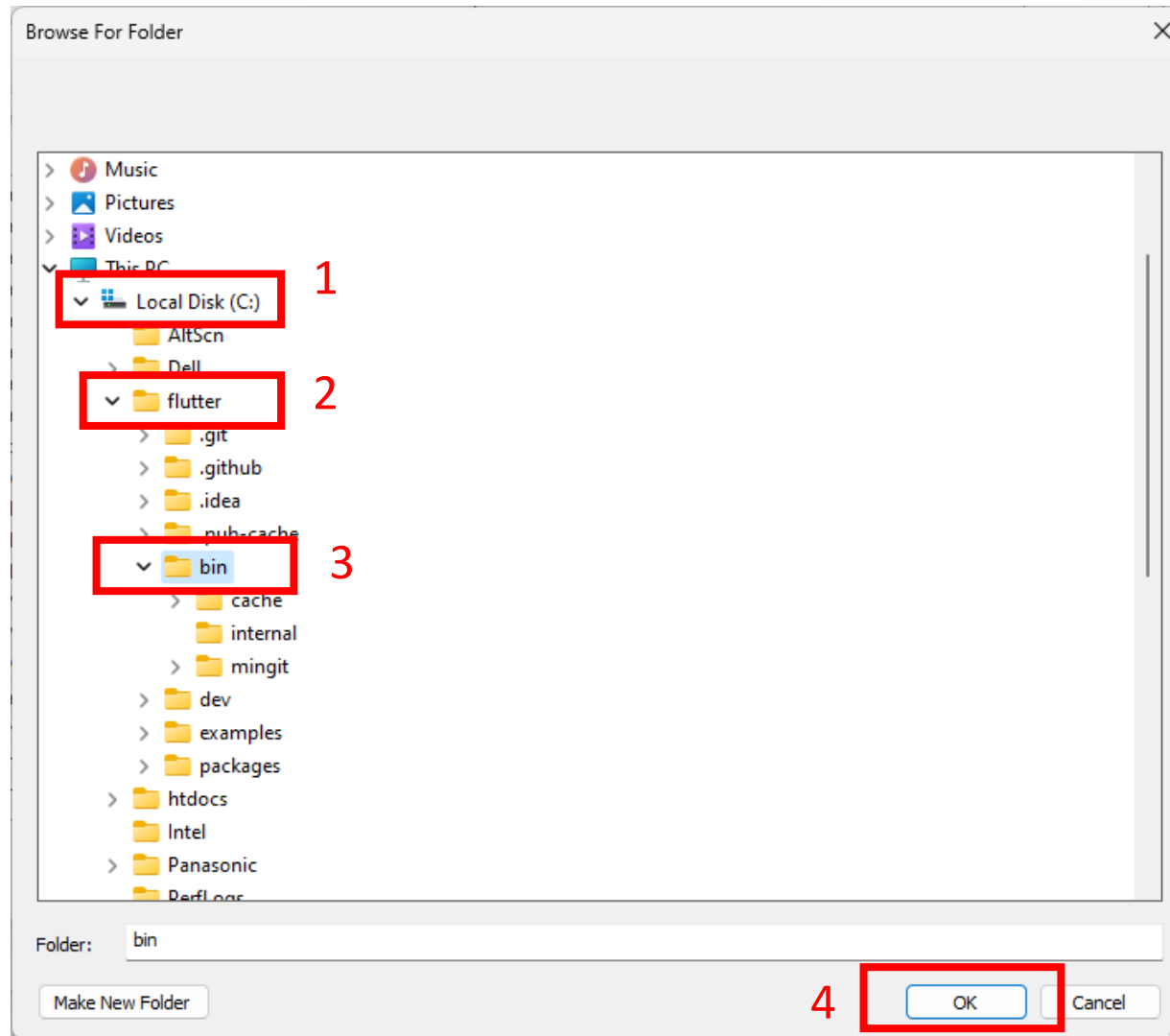
4. Instalasi Flutter SDK



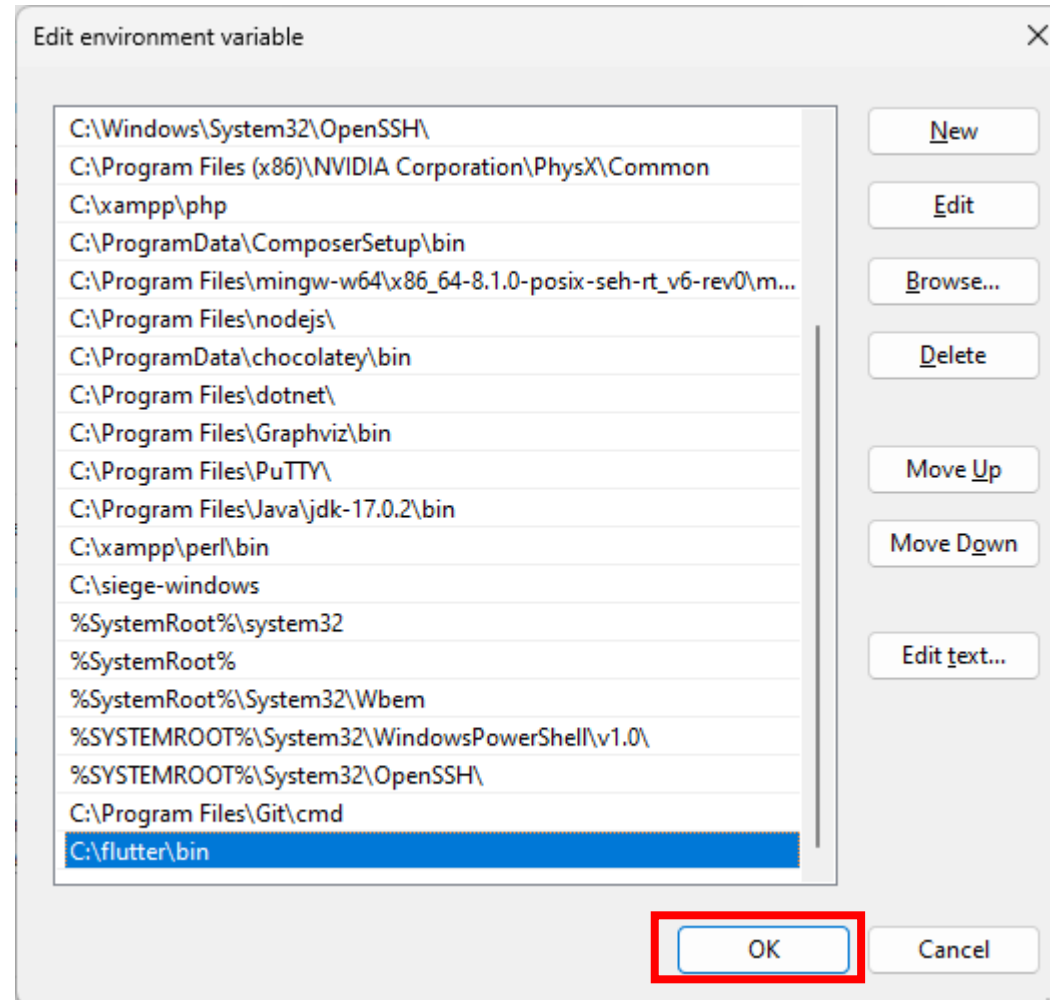
4. Instalasi Flutter SDK



4. Instalasi Flutter SDK

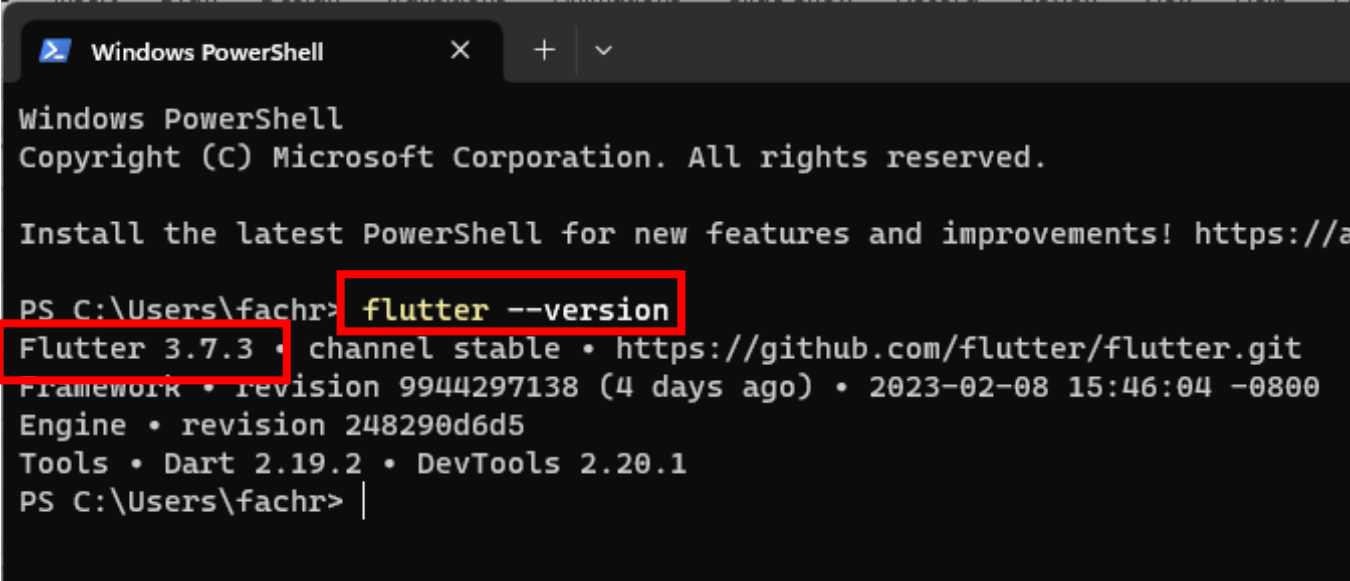


4. Instalasi Flutter SDK



4. Instalasi Flutter SDK

- Jalankan perintah **flutter --version** pada terminal
- Jika flutter berhasil diatur ke dalam environment variable, maka versi flutter akan ditampilkan



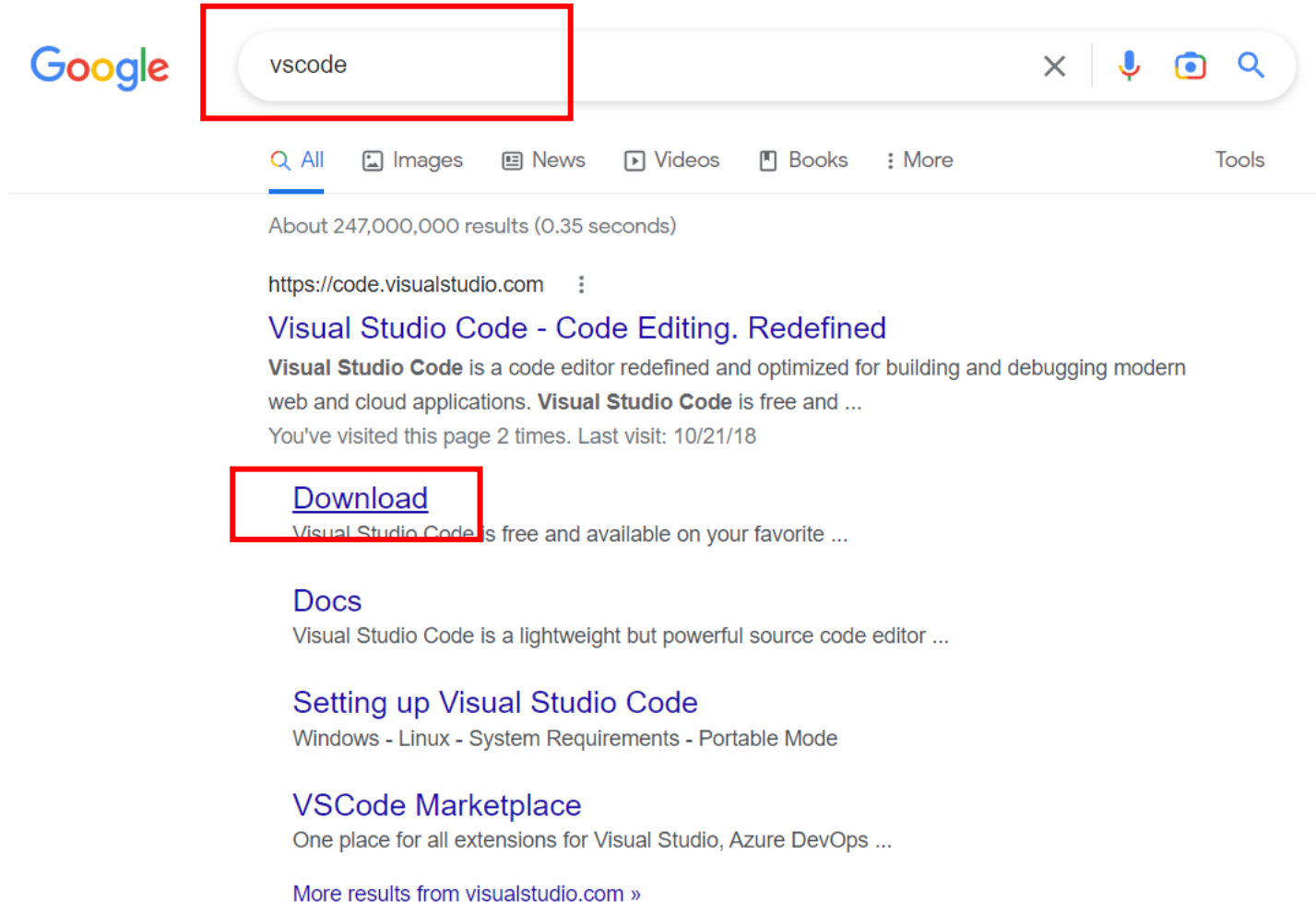
```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PowerShellLatest

PS C:\Users\fachr> flutter --version
Flutter 3.7.3 • channel stable • https://github.com/flutter/flutter.git
Framework • revision 9944297138 (4 days ago) • 2023-02-08 15:46:04 -0800
Engine • revision 248290d6d5
Tools • Dart 2.19.2 • DevTools 2.20.1
PS C:\Users\fachr> |
```

5. Instalasi VSCode

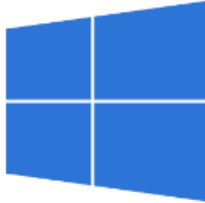
5. Instalasi VSCode



5. Instalasi VSCode

Download Visual Studio Code

Free and built on open source. Integrated Git, debugging and extensions.



 **Windows**
Windows 8, 10, 11

User Installer	x64	x86	Arm64
System Installer	x64	x86	Arm64
.zip	x64	x86	Arm64
CLI	x64	x86	Arm64



 **.deb**
Debian, Ubuntu

 **.rpm**
Red Hat, Fedora, SUSE

.deb	x64	Arm32	Arm64
.rpm	x64	Arm32	Arm64
.tar.gz	x64	Arm32	Arm64
Snap	Snap Store		
CLI	x64	Arm32	Arm64



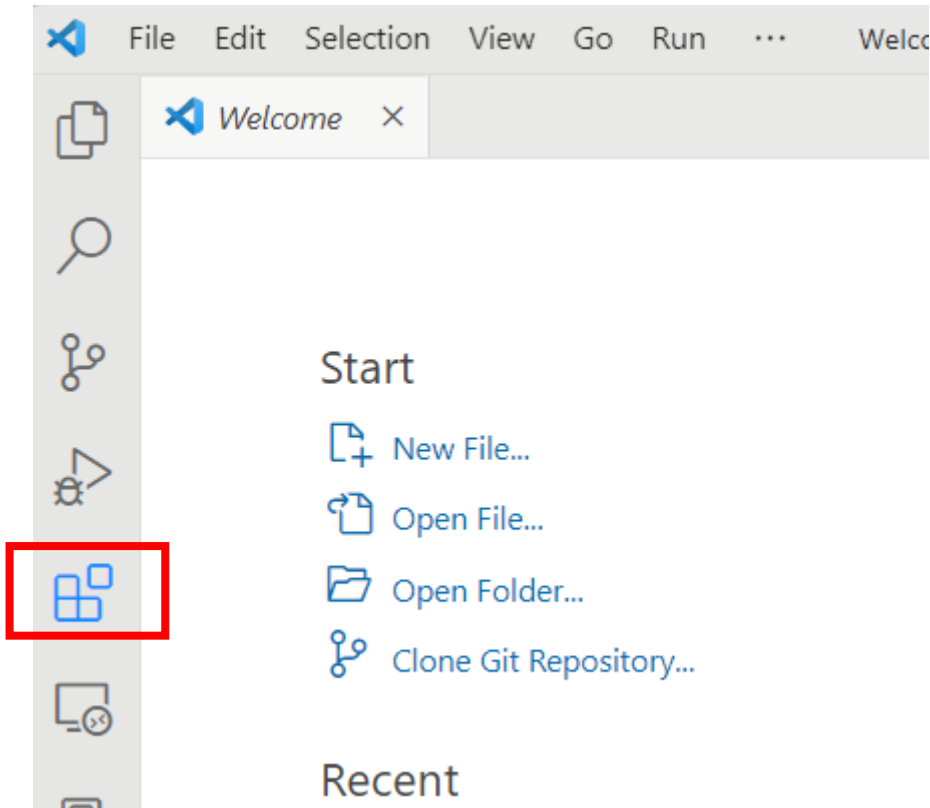
 **Mac**
macOS 10.11+

.zip	Intel chip	Apple silicon	Universal
CLI	Intel chip	Apple silicon	

6. Instalasi Extensions VSCode (Dart dan Flutter)

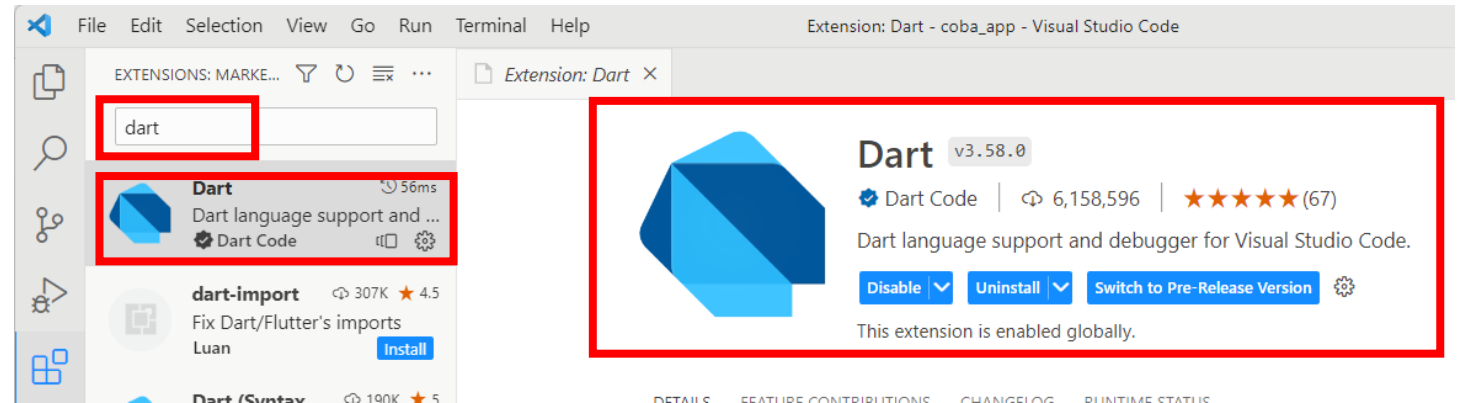
6. Instalasi Extension VSCode (Dart & Flutter)

- Buka VSCode
- Klik icon extensions

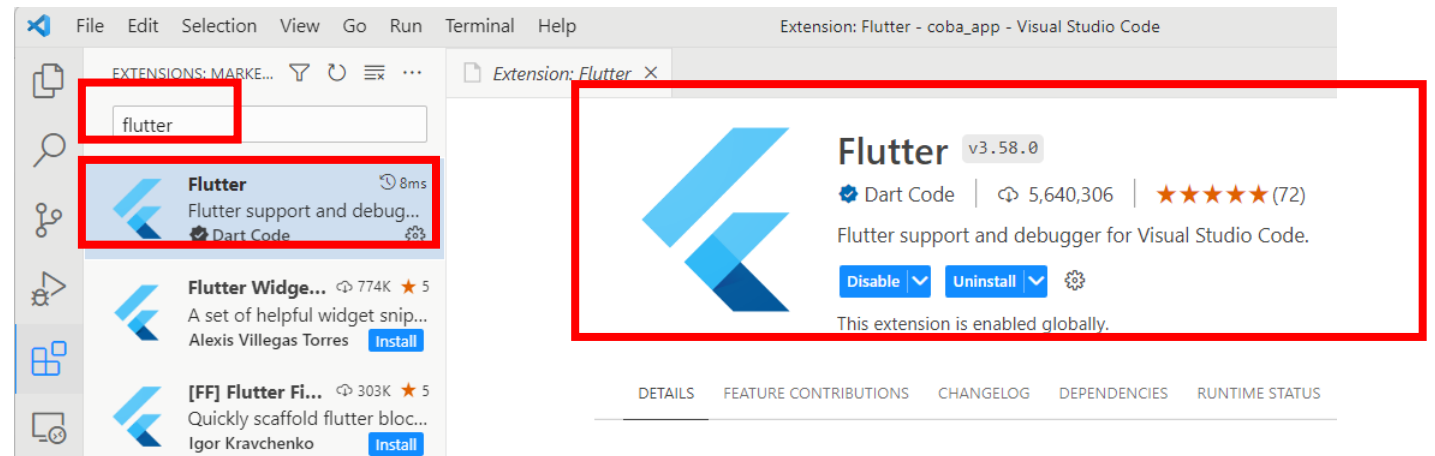


6. Instalasi Extension VSCode (Dart & Flutter)

- Cari dart → install



- Cari flutter → install



7. Instalasi Android SDK

7. Instalasi Android SDK

- a. Download cmdline-tools.zip
- b. Extract cmdline-tools.zip
- c. Move folder cmdline-tools ke C:\
- d. Masuk ke folder bin di dalam folder cmdline-tools
- e. Buka cmd dari folder bin
- f. Install build tools
- g. Install platform android
- h. Install cmdline-tools
- i. Set environment variable
- j. Periksa status dengan flutter doctor
- k. Install android licenses

7. Instalasi Android SDK

- Sampai pada tahap ini jika perintah **flutter doctor** dieksekusi, maka ditampilkan informasi sebagai berikut:

```
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

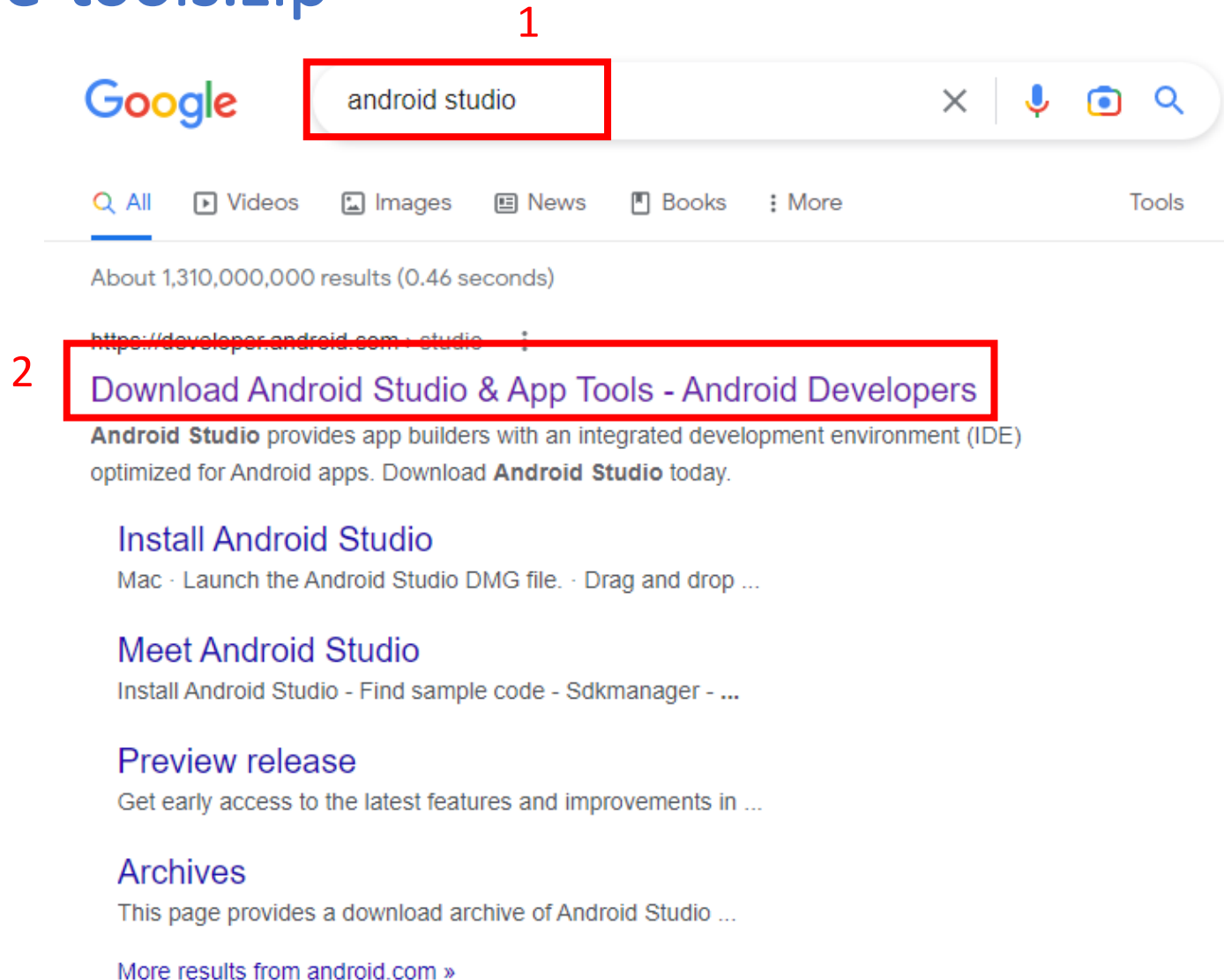
C:\Users\fach>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.7.3, on Microsoft Windows [Version 10.0.22621.1265], locale en-ID)
[✓] Windows Version (Installed version of Windows is version 10 or higher)
[!] Android toolchain - develop for Android devices (Android SDK version 30.0.3)
    X cmdline-tools component is missing
      Run `path/to/sdkmanager --install "cmdline-tools;latest"`
      See https://developer.android.com/studio/command-line for more details.
    X Android license status unknown.
      Run `flutter doctor --android-licenses` to accept the SDK licenses.
      See https://flutter.dev/docs/get-started/install/windows#android-setup for more details.
[✓] Chrome - develop for the web
[!] Visual Studio - develop for Windows
    X Visual Studio not installed; this is necessary for Windows development.
      Download at https://visualstudio.microsoft.com/downloads/.
      Please install the "Desktop development with C++" workload, including all of its default components
[!] Android Studio (not installed)
[✓] VS Code (version 1.75.1)
[✓] Connected device (3 available)
[✓] HTTP Host Availability

! Doctor found issues in 3 categories.
```

7. Instalasi Android SDK

a. Download cmdline-tools.zip

- Cari android studio pada google



7. Instalasi Android SDK

a. Download cmdline-tools.zip

Ketik **ctrl + F**

developers 

Platform

Android Studio

Google Play

Jetpack

More ▾

command|

1/6



ish ▾

Si

1

Command line tools only

Platform	SDK tools package	Size	SHA-256 checksum
Windows	command linetools-win-9477386_latest.zip	133.5 MB	696431978daadd33a28841320659835ba8db8080a535b8f35e9e60701ab8b491
Mac	command linetools-mac-9477386_latest.zip	133.5 MB	2072ffce4f54cdc0e6d2074d2f381e7e579b7d63e915c220b96a7db95b2900ee
Linux	command linetools-linux-9477386_latest.zip	133.5 MB	bd1aa17c7ef10066949c88dc6c9c8d536be27f992a1f3b5a584f9bd2ba5646a0

2

Command-line tools are included in Android Studio. If you do not need Android Studio, you can download the basic Android **command**-line tools above. You can use the included [sdkmanager](#) to download other SDK packages.

7. Instalasi Android SDK

b. Extract cmdline-tools.zip

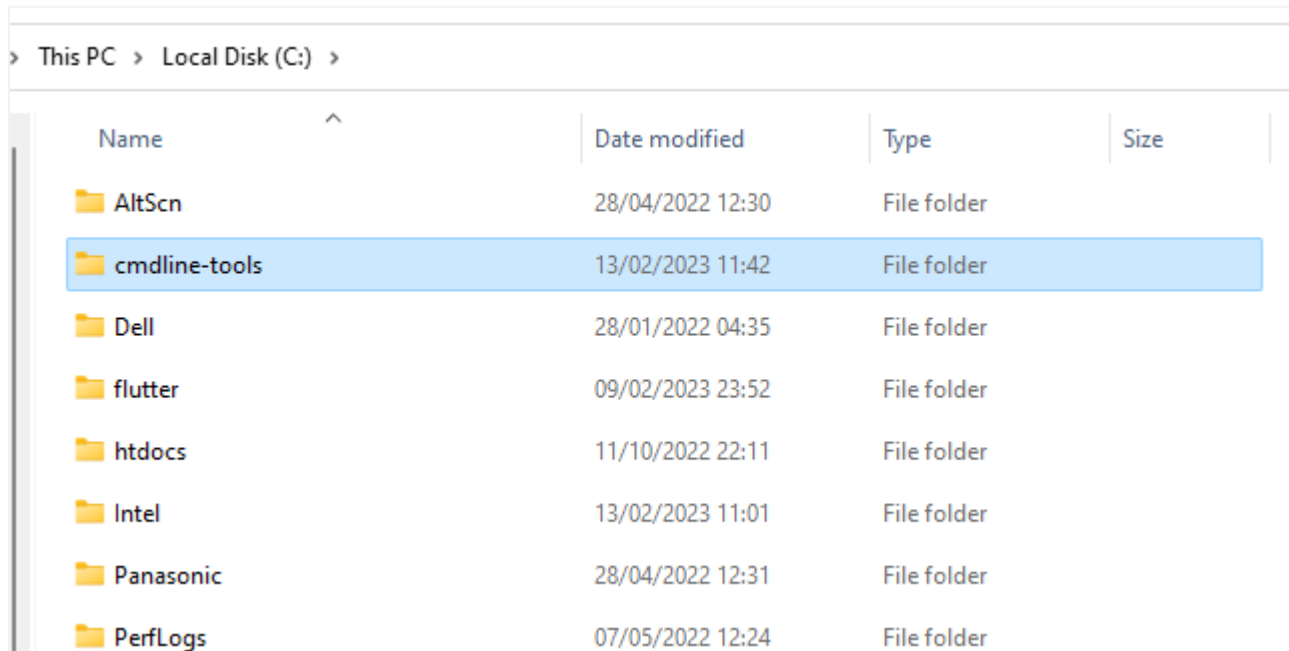
- Extract file *.zip

Name	Date modified
cmdline-tools	13/02/2023 11:42
commandlinetools-win-9477386_latest.zip	08/02/2023 11:11
flutter_windows_3.7.3-stable.zip	13/02/2023 10:54
jdk-17_windows-x64_bin.exe	08/02/2023 11:29
VSCodeUserSetup-x64-1.75.0.exe	08/02/2023 11:33

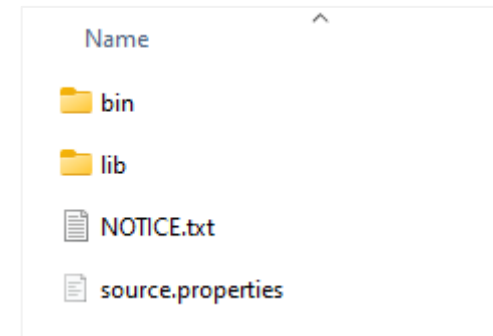
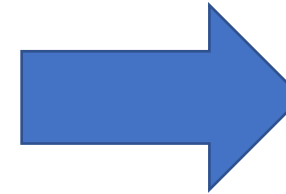
7. Instalasi Android SDK

c. Move folder cmdline-tools ke C:\

- Pindahkan folder hasil extract ke C:\



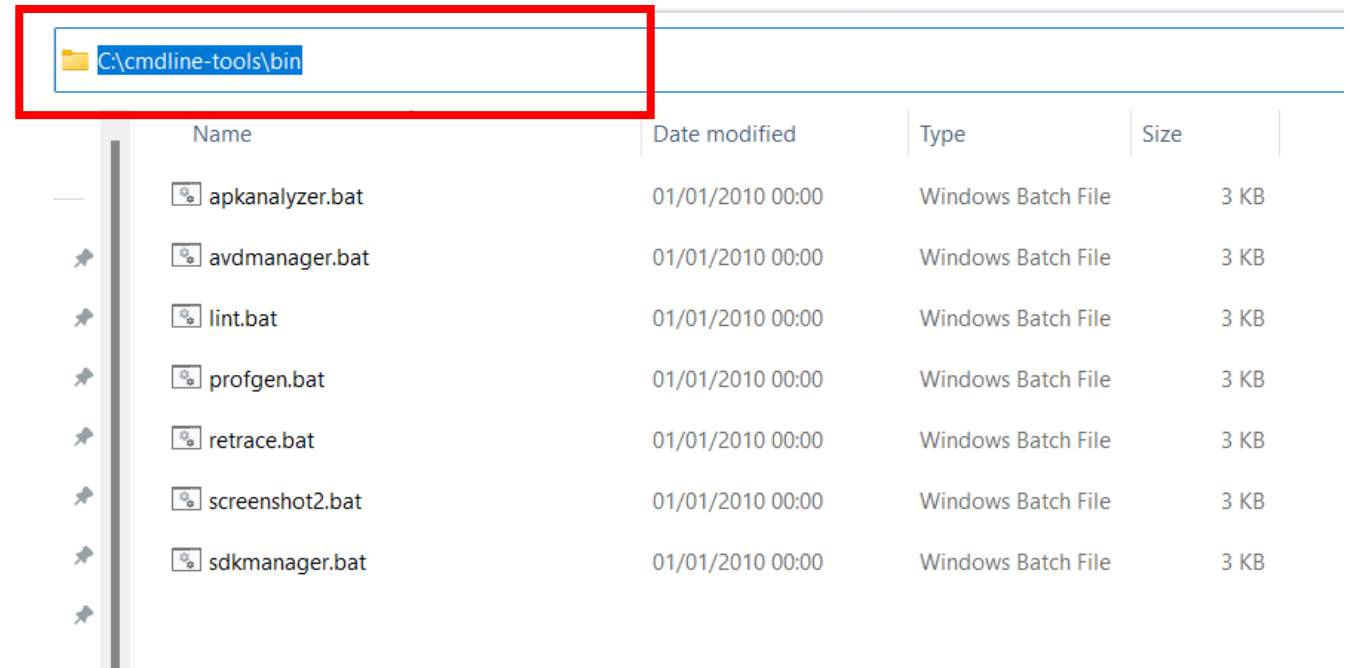
folder cmdline-tools berisi:



7. Instalasi Android SDK

d. Masuk ke folder bin di dalam folder cmdline-tools

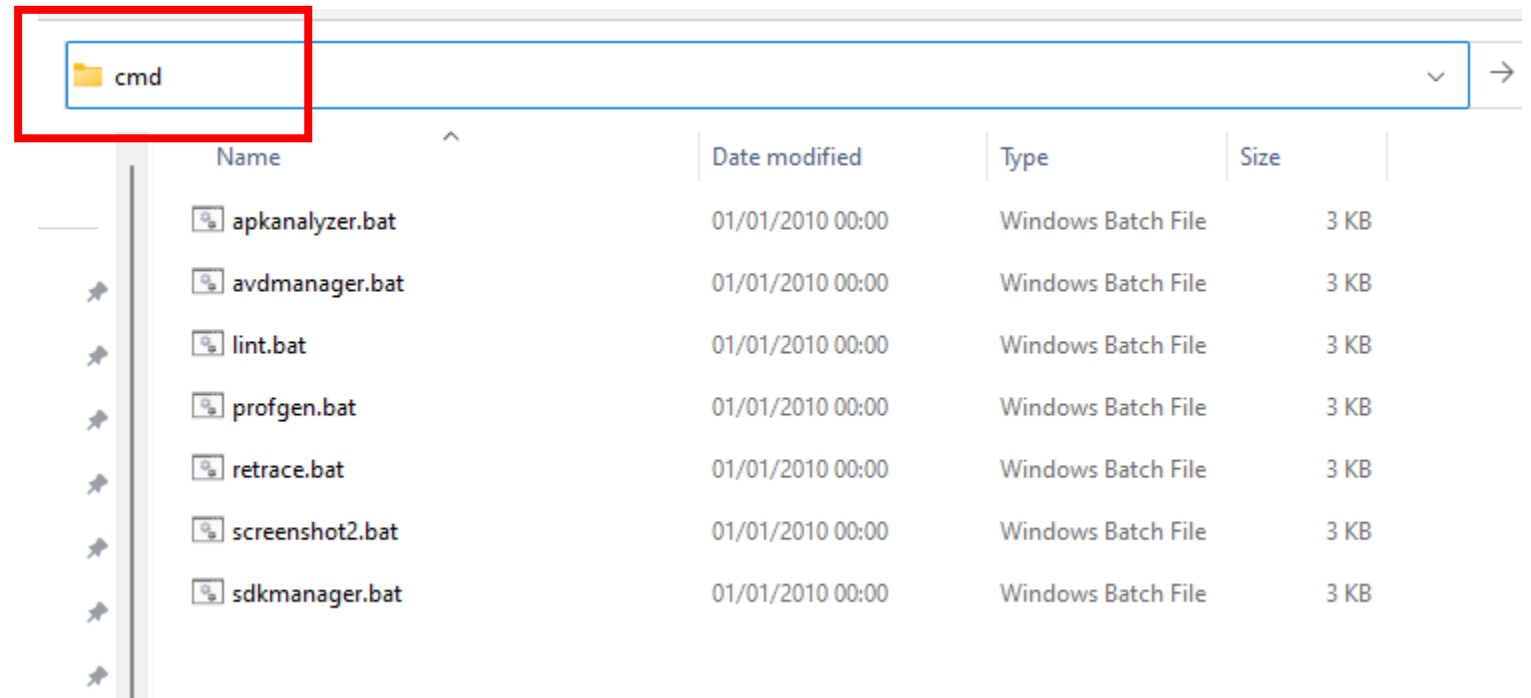
- Masuk ke folder bin di dalam folder cmdline-tools
- Klik address bar pada explorer



7. Instalasi Android SDK

e. Buka cmd dari folder bin

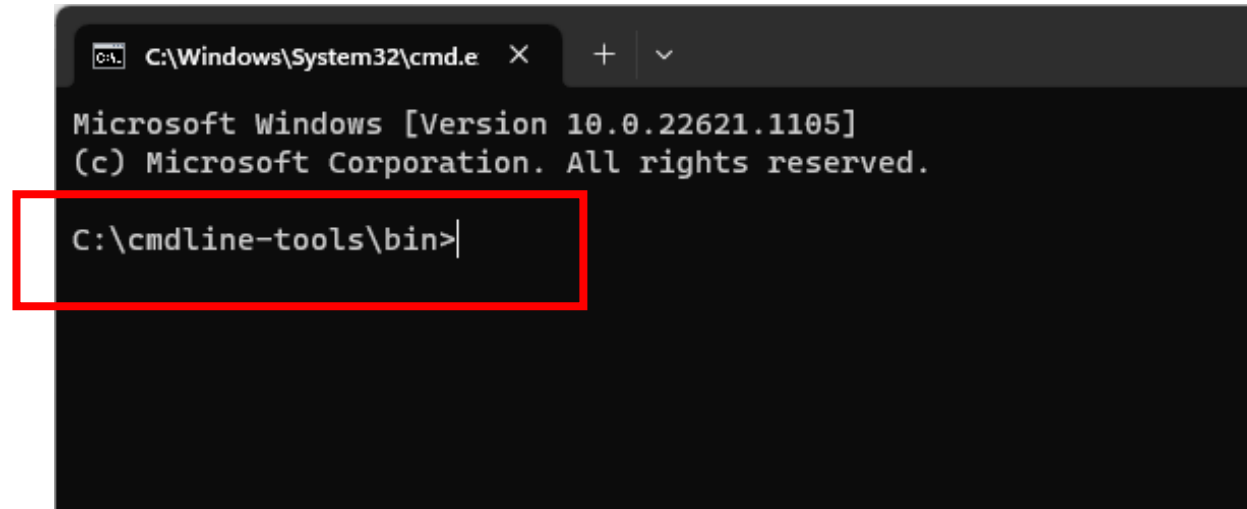
- Ubah tulisan menjadi cmd
- Tekan Enter



7. Instalasi Android SDK

e. Buka cmd dari folder bin

- Muncul command prompt dengan lokasi path **C:\cmdline-tools\bin**

A screenshot of a Windows Command Prompt window. The title bar shows 'C:\Windows\System32\cmd.e' with a close button. The window content displays 'Microsoft Windows [Version 10.0.22621.1105]' and '(c) Microsoft Corporation. All rights reserved.' Below this, the current directory is 'C:\cmdline-tools\bin>', which is highlighted by a red rectangular box. The prompt is followed by a vertical cursor line.

```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.22621.1105]
(c) Microsoft Corporation. All rights reserved.
C:\cmdline-tools\bin>|
```

7. Instalasi Android SDK

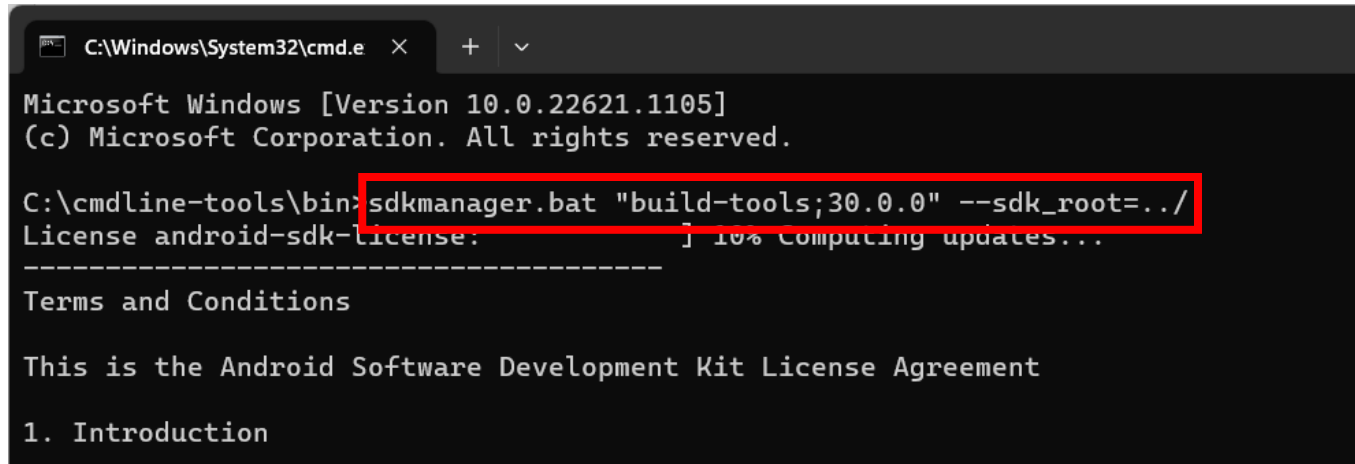
f. Install build tools

- **Build tools** adalah komponen dari android SDK yang dibutuhkan untuk membangun Aplikasi Android
- Komponen build tools perlu untuk selalu diperbarui

7. Instalasi Android SDK

f. Install build tools

- Instalasi build tools dilakukan dengan perintah berikut:
- `sdkmanager.bat "build-tools;30.0.0" --sdk_root=../`



```
C:\Windows\System32\cmd.e  X  +  v
Microsoft Windows [Version 10.0.22621.1105]
(c) Microsoft Corporation. All rights reserved.

C:\cmdline-tools\bin> sdkmanager.bat "build-tools;30.0.0" --sdk_root=../
License android-sdk-license: 10% Computing updates...
-----
Terms and Conditions

This is the Android Software Development Kit License Agreement

1. Introduction
```

7. Instalasi Android SDK

f. Install build tools

- Informasi versi build tools terbaru:

<https://developer.android.com/studio/command-line/sdkmanager>

Install packages

To install packages, use the following syntax:

```
sdkmanager packages [options]
```

The *packages* argument is an SDK-style path, as shown with the `--list` command, wrapped in quotes. For example, `"build-tools;33.0.1"` or `"platforms;android-33"`.

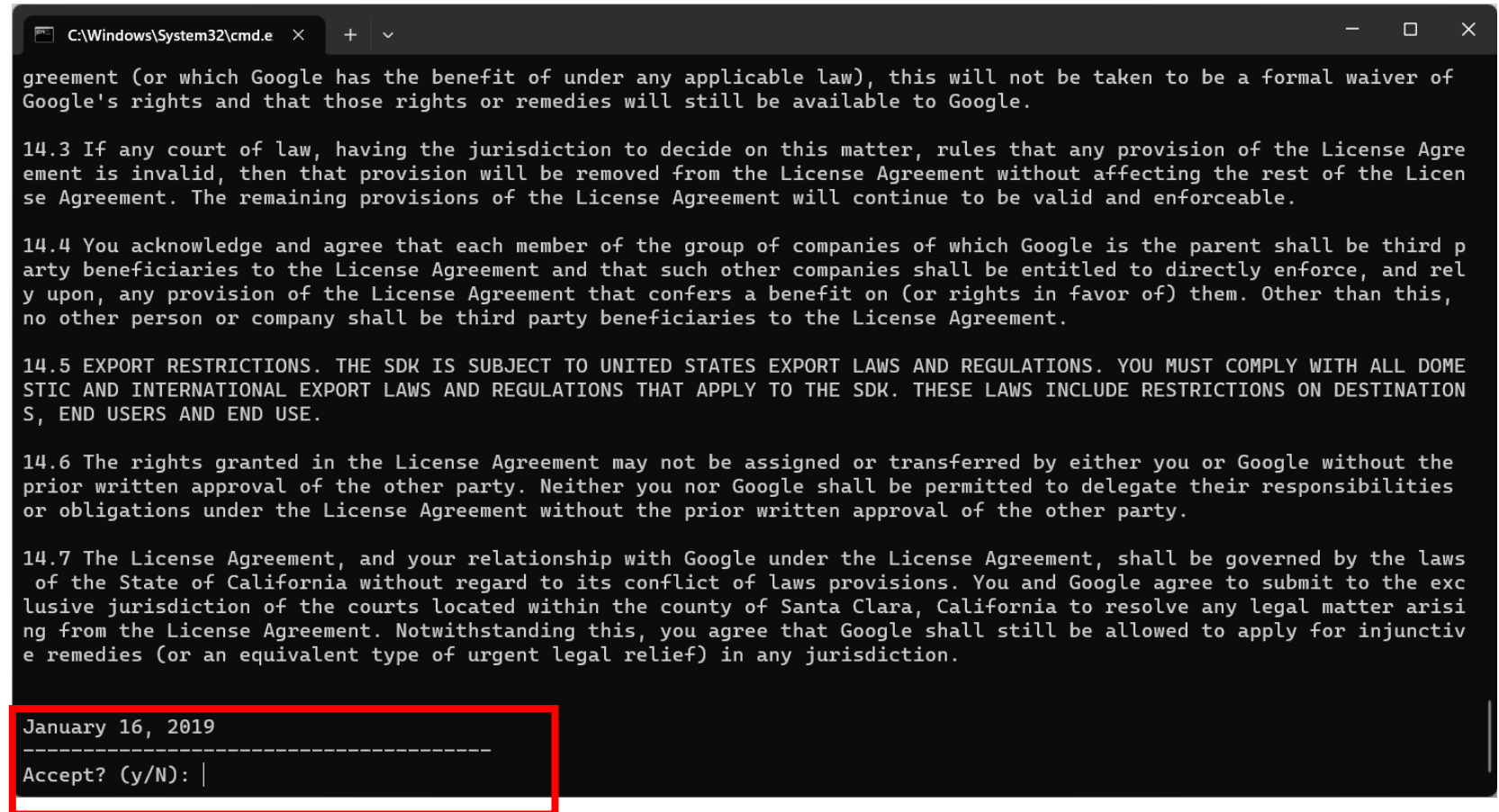
You can pass multiple package paths, separated with a space, but they must each be wrapped in their own set of quotes. For example, here's how to install the latest platform tools and the SDK tools for API level 33:

```
sdkmanager "platform-tools" "platforms;android-33"
```

7. Instalasi Android SDK

f. Install build tools

- Ketik **y** pada pertanyaan berikut



```
C:\Windows\System32\cmd.e  +  v

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STIC AND INTERNATIONAL EXPORT LAWS AND REGULATIONS THAT APPLY TO THE SDK. THESE LAWS INCLUDE RESTRICTIONS ON DESTINATION
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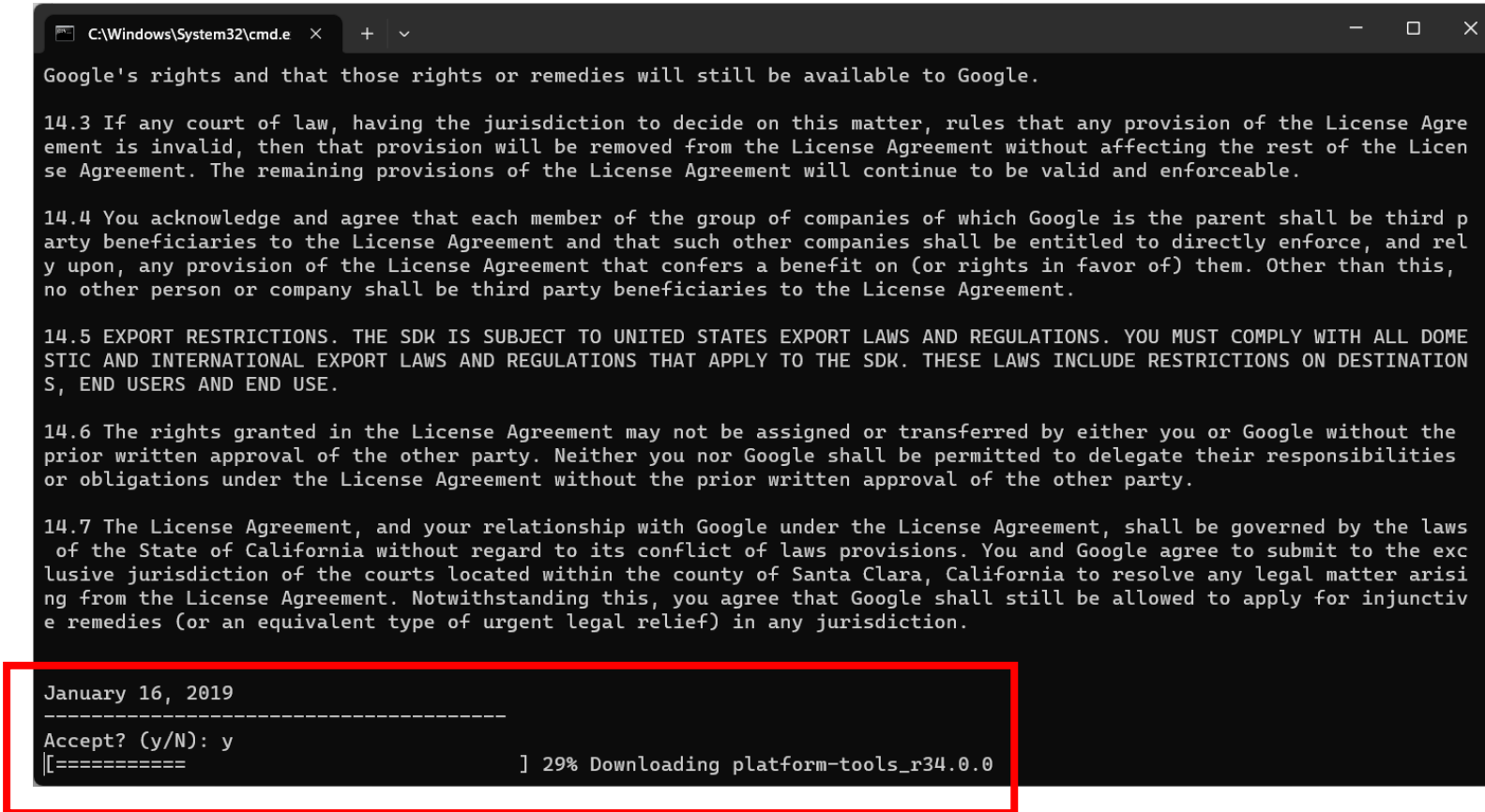
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January 16, 2019
-----
Accept? (y/N): |
```

7. Instalasi Android SDK

f. Install build tools



```
C:\Windows\System32\cmd.e

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14.5 EXPORT RESTRICTIONS. THE SDK IS SUBJECT TO UNITED STATES EXPORT LAWS AND REGULATIONS. YOU MUST COMPLY WITH ALL DOMESTIC AND INTERNATIONAL EXPORT LAWS AND REGULATIONS THAT APPLY TO THE SDK. THESE LAWS INCLUDE RESTRICTIONS ON DESTINATIONS, END USERS AND END USE.

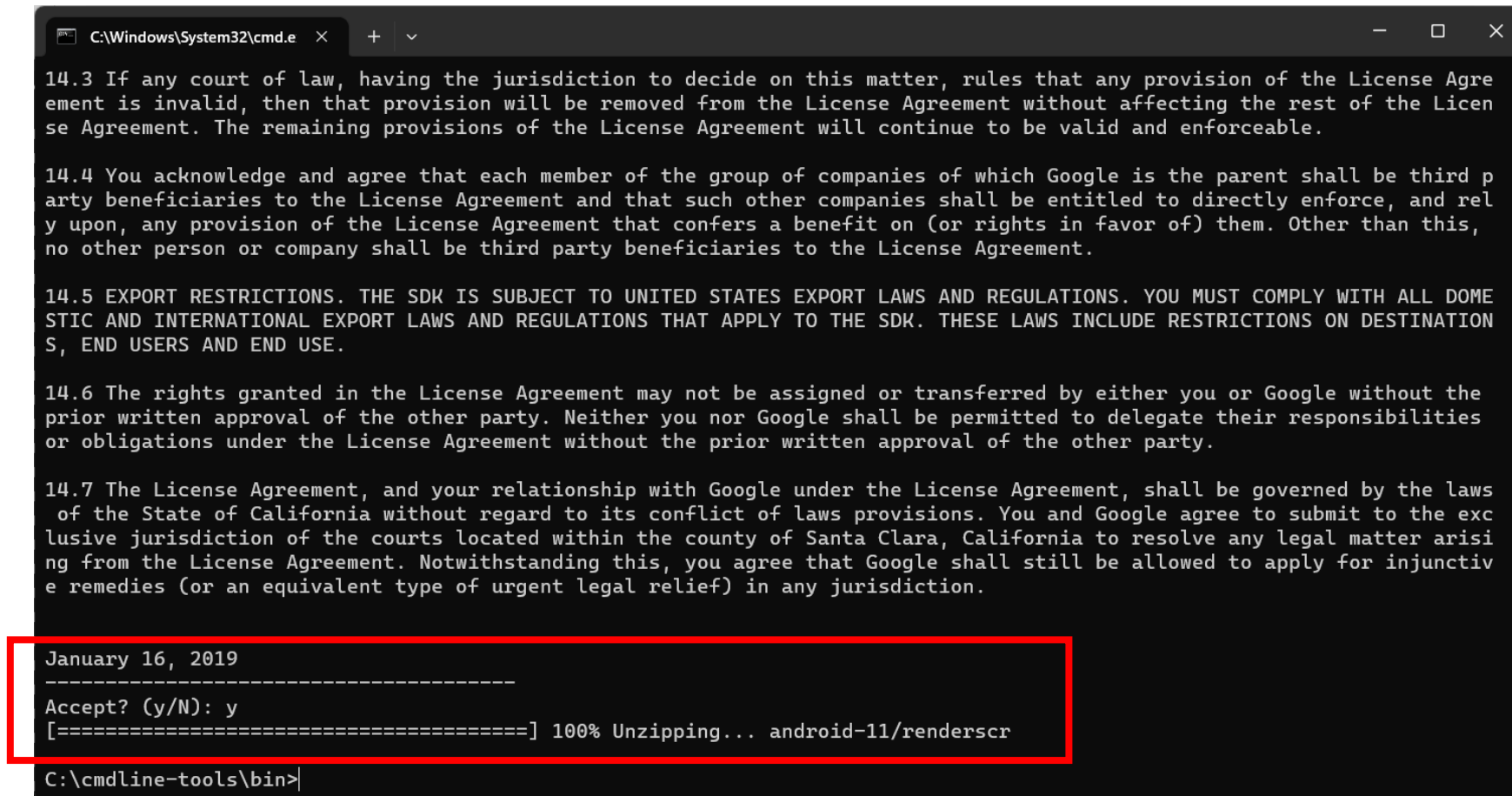
14.6 The rights granted in the License Agreement may not be assigned or transferred by either you or Google without the prior written approval of the other party. Neither you nor Google shall be permitted to delegate their responsibilities or obligations under the License Agreement without the prior written approval of the other party.

14.7 The License Agreement, and your relationship with Google under the License Agreement, shall be governed by the laws of the State of California without regard to its conflict of laws provisions. You and Google agree to submit to the exclusive jurisdiction of the courts located within the county of Santa Clara, California to resolve any legal matter arising from the License Agreement. Notwithstanding this, you agree that Google shall still be allowed to apply for injunctive remedies (or an equivalent type of urgent legal relief) in any jurisdiction.

January 16, 2019
-----
Accept? (y/N): y
[===== ] 29% Downloading platform-tools_r34.0.0
```


7. Instalasi Android SDK

f. Install build tools



```
C:\Windows\System32\cmd.e
14.3 If any court of law, having the jurisdiction to decide on this matter, rules that any provision of the License Agreement is invalid, then that provision will be removed from the License Agreement without affecting the rest of the License Agreement. The remaining provisions of the License Agreement will continue to be valid and enforceable.

14.4 You acknowledge and agree that each member of the group of companies of which Google is the parent shall be third party beneficiaries to the License Agreement and that such other companies shall be entitled to directly enforce, and rely upon, any provision of the License Agreement that confers a benefit on (or rights in favor of) them. Other than this, no other person or company shall be third party beneficiaries to the License Agreement.

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14.6 The rights granted in the License Agreement may not be assigned or transferred by either you or Google without the prior written approval of the other party. Neither you nor Google shall be permitted to delegate their responsibilities or obligations under the License Agreement without the prior written approval of the other party.

14.7 The License Agreement, and your relationship with Google under the License Agreement, shall be governed by the laws of the State of California without regard to its conflict of laws provisions. You and Google agree to submit to the exclusive jurisdiction of the courts located within the county of Santa Clara, California to resolve any legal matter arising from the License Agreement. Notwithstanding this, you agree that Google shall still be allowed to apply for injunctive remedies (or an equivalent type of urgent legal relief) in any jurisdiction.

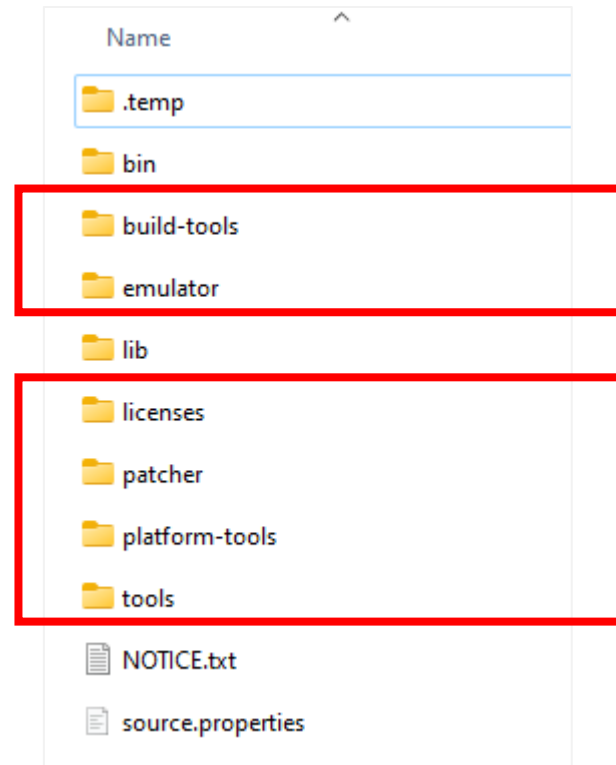
January 16, 2019
-----
Accept? (y/N): y
[=====] 100% Unzipping... android-11/renderscr

C:\cmdline-tools\bin>
```

7. Instalasi Android SDK

f. Install build tools

- Berikut isi folder cmdline-tools setelah install build tools



7. Instalasi Android SDK

g. Install platform android

- **Platform android** adalah komponen yang menentukan versi dari suatu platform android yang akan dituju
 - aplikasi di-compile ke versi android tersebut
- Misalnya
 - Android 11 (API level 30)
 - Android 12 (API level 31, 32)
 - Android 13 (API level 33)
 - Android 14 (Developer Preview) ← #15Feb2023

**<https://developer.android.com/studio/releases/platforms>*

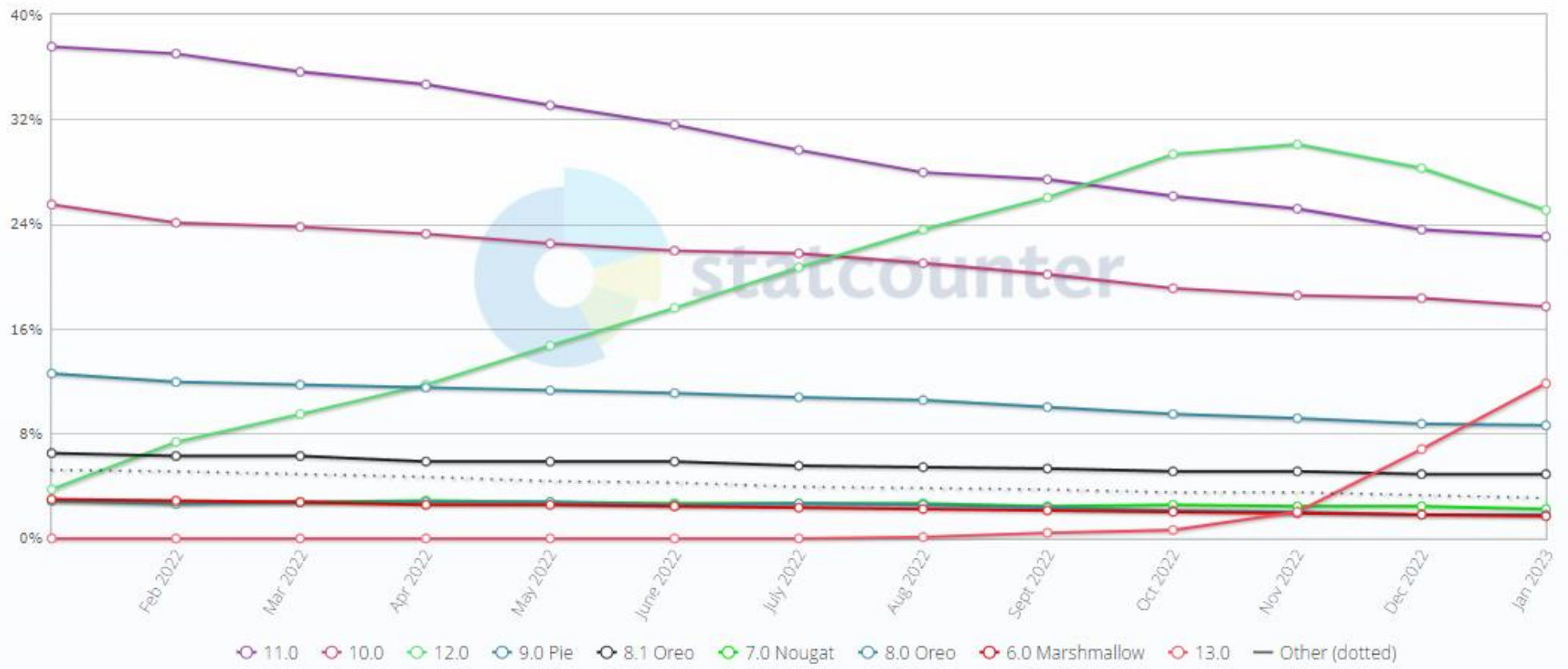
mo



Android Version Market Share Worldwide

Jan 2022 - Jan 2023

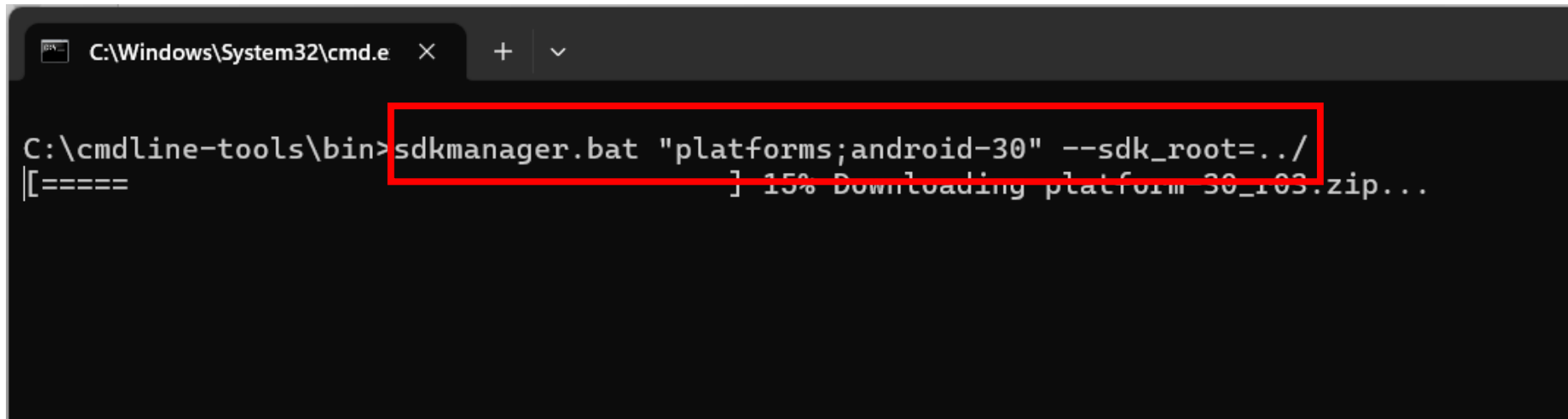
Edit Chart Data



7. Instalasi Android SDK

g. Install platform android

- Instalasi platform android dilakukan dengan perintah berikut:
- `sdkmanager.bat "platforms;android-30" --sdk_root=../`

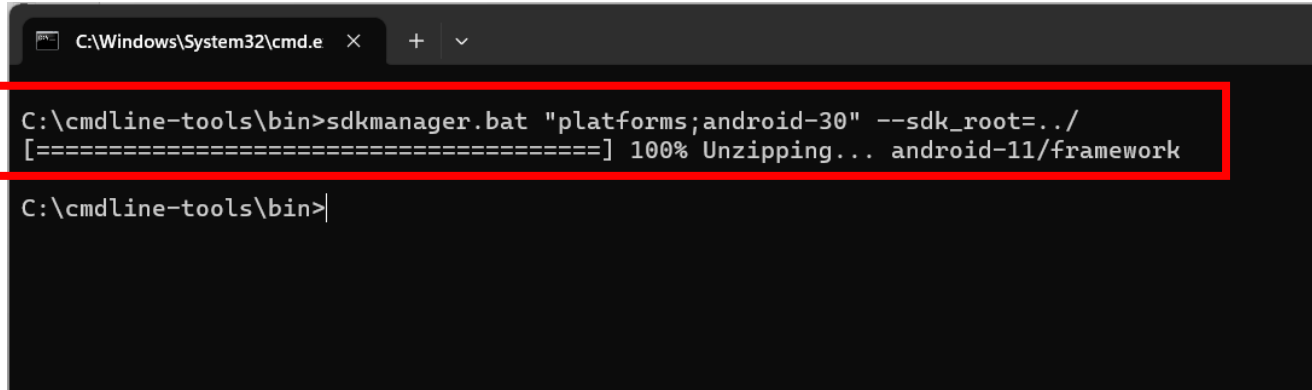


```
C:\Windows\System32\cmd.e  x  +  v

C:\cmdline-tools\bin>sdkmanager.bat "platforms;android-30" --sdk_root=../
[=====] 15% Downloading platform 30_r03.zip...
```

7. Instalasi Android SDK

g. Install platform android

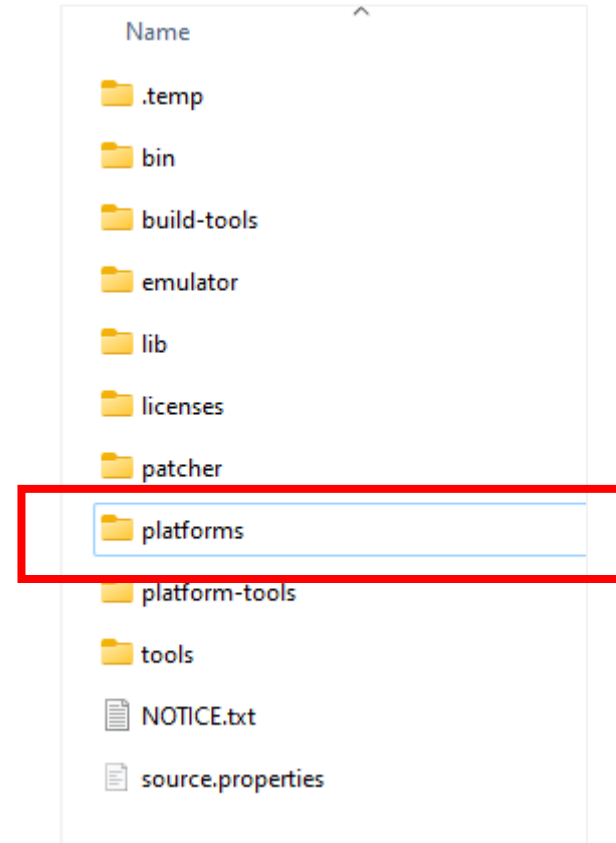


```
C:\Windows\System32\cmd.e  X  +  v  
C:\cmdline-tools\bin>sdkmanager.bat "platforms;android-30" --sdk_root=../  
[=====] 100% Unzipping... android-11/framework  
C:\cmdline-tools\bin>
```

7. Instalasi Android SDK

g. Install platform android

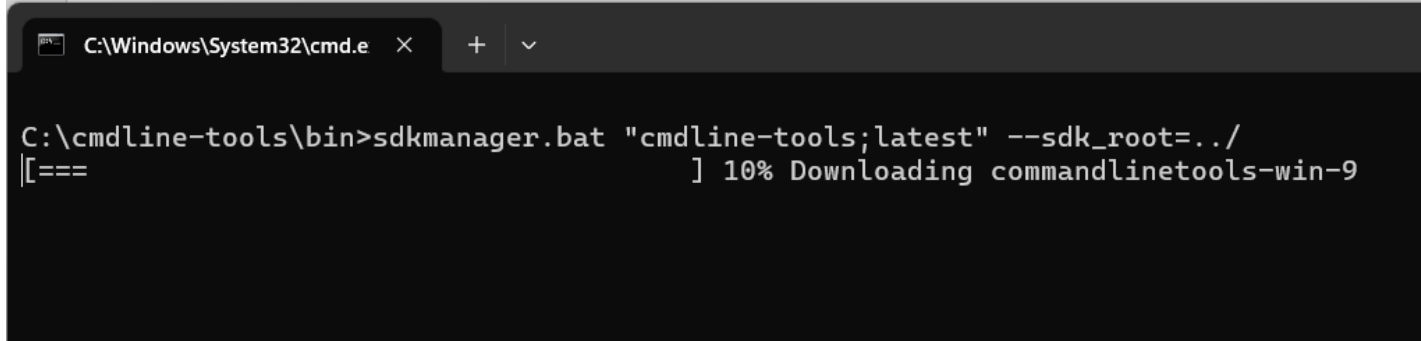
- Berikut isi folder cmdline-tools setelah install platform android



7. Instalasi Android SDK

h. Install cmdline tools

- Ketik perintah berikut:
- `sdkmanager.bat "cmdline-tools;latest" --sdk_root=../`

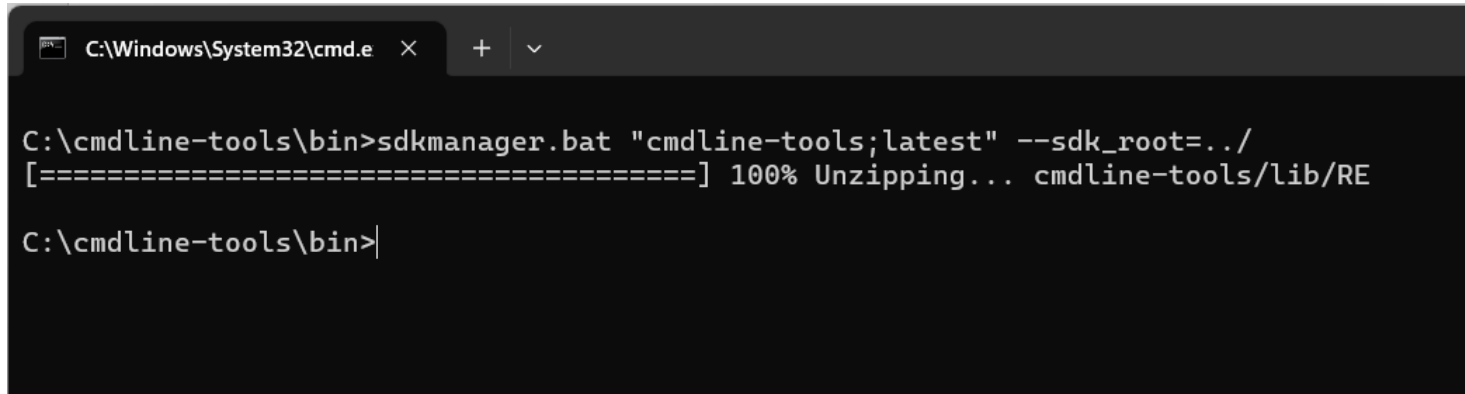


```
C:\Windows\System32\cmd.e  x  +  v

C:\cmdline-tools\bin>sdkmanager.bat "cmdline-tools;latest" --sdk_root=../
|[===                               ] 10% Downloading commandlinetools-win-9
```


7. Instalasi Android SDK

h. Install cmdline tools

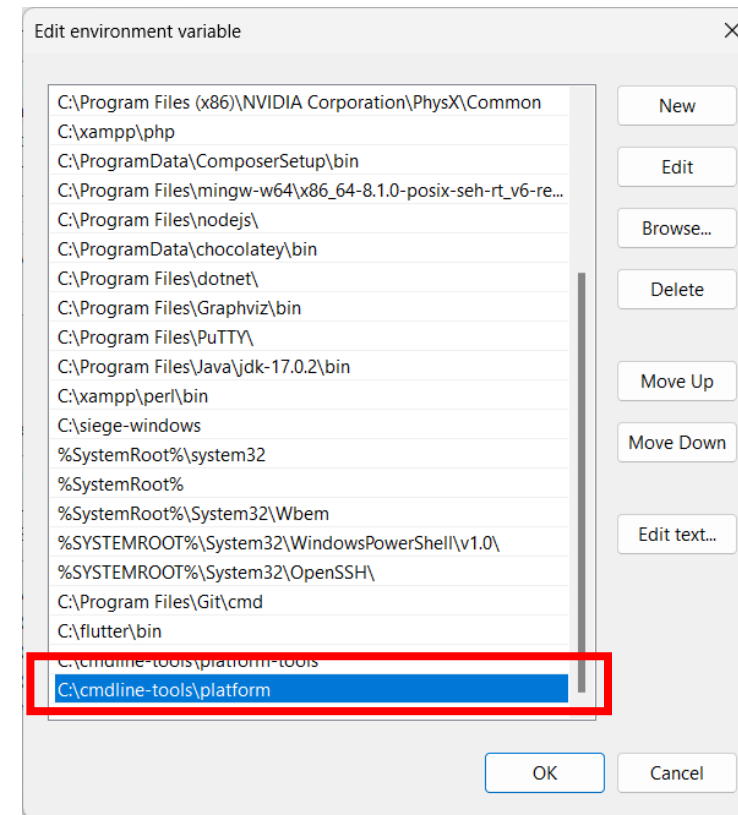
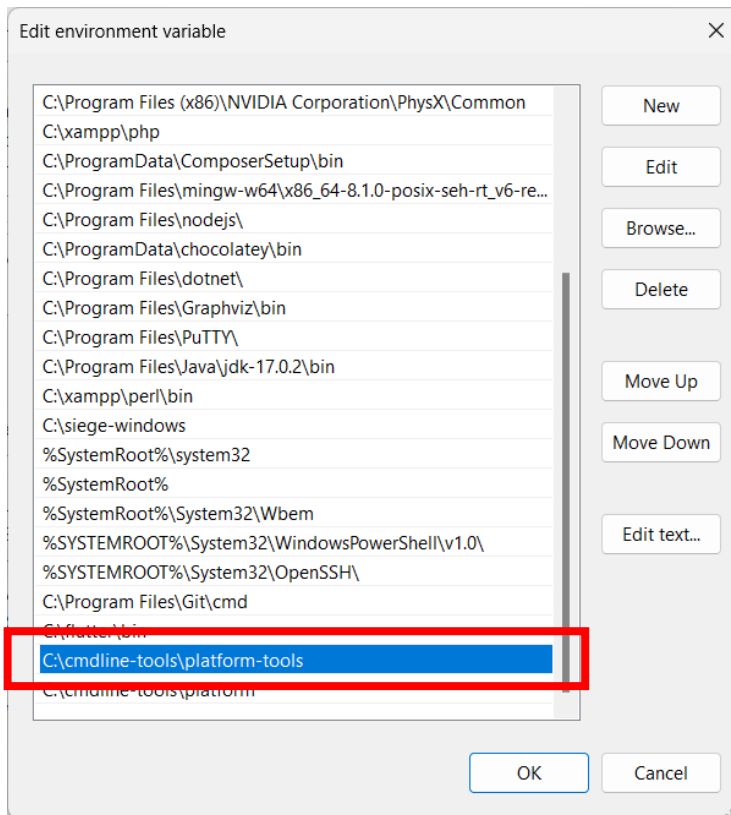


```
C:\Windows\System32\cmd.e  ×  +  v  
  
C:\cmdline-tools\bin>sdkmanager.bat "cmdline-tools;latest" --sdk_root=../  
[=====] 100% Unzipping... cmdline-tools/lib/RE  
  
C:\cmdline-tools\bin>
```

7. Instalasi Android SDK

i. Set environment variable

- Tambahkan path ***C:\cmdline-tools\platform-tools*** dan ***C:\cmdline-tools\platform-tools*** pada environment variable



7. Instalasi Android SDK

j. Periksa status dengan flutter doctor

- Ketik perintah `flutter doctor`

instalasi berhasil ☒

```
C:\Windows\System32\cmd.e x + v
C:\cmdline-tools\bin>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.7.3, on Microsoft Windows [Version 10.0.22H2], locale en-ID)
[✓] Windows Version (Installed version of Windows is version 10 or higher)
[!] Android toolchain - develop for Android devices (Android SDK version 30.0.3)
    ! Some Android licenses not accepted. To resolve this, run: flutter doctor --android-licenses
[✓] Chrome - develop for the web
[✓] Visual Studio - develop for Windows (Visual Studio Build Tools 2019 16.11.9)
[!] Android Studio (not installed)
[✓] VS Code (version 1.75.1)
[✓] Connected device (3 available)
[✓] HTTP Host Availability

! Doctor found issues in 2 categories.
C:\cmdline-tools\bin>
```

instalasi gagal ✗

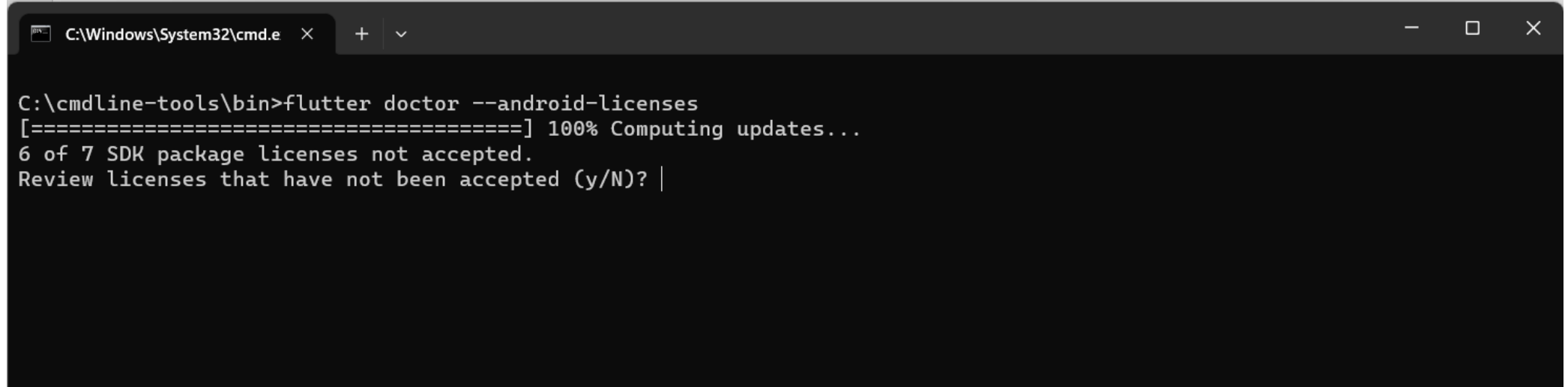
```
C:\Windows\System32\cmd.e x + v
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.
C:\cmdline-tools\bin>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.7.3, on Microsoft Windows [Version 10.0.22621.1265], locale en-ID)
[✓] Windows Version (Installed version of Windows is version 10 or higher)
[!] Android toolchain - develop for Android devices (Android SDK version 30.0.3)
    ✗ cmdline-tools component is missing
      Run 'path/to/sdkmanager --install "cmdline-tools;latest"'
      See https://developer.android.com/studio/command-line for more details.
    ✗ Android license status unknown.
      Run 'flutter doctor --android-licenses' to accept the SDK licenses.
      See https://flutter.dev/docs/get-started/install/windows#android-setup for more details.
[✓] Chrome - develop for the web
[✓] Visual Studio - develop for Windows (Visual Studio Build Tools 2019 16.11.9)
[!] Android Studio (not installed)
[✓] VS Code (version 1.75.1)
[✓] Connected device (3 available)
[✓] HTTP Host Availability

! Doctor found issues in 2 categories.
C:\cmdline-tools\bin>
```

7. Instalasi Android SDK

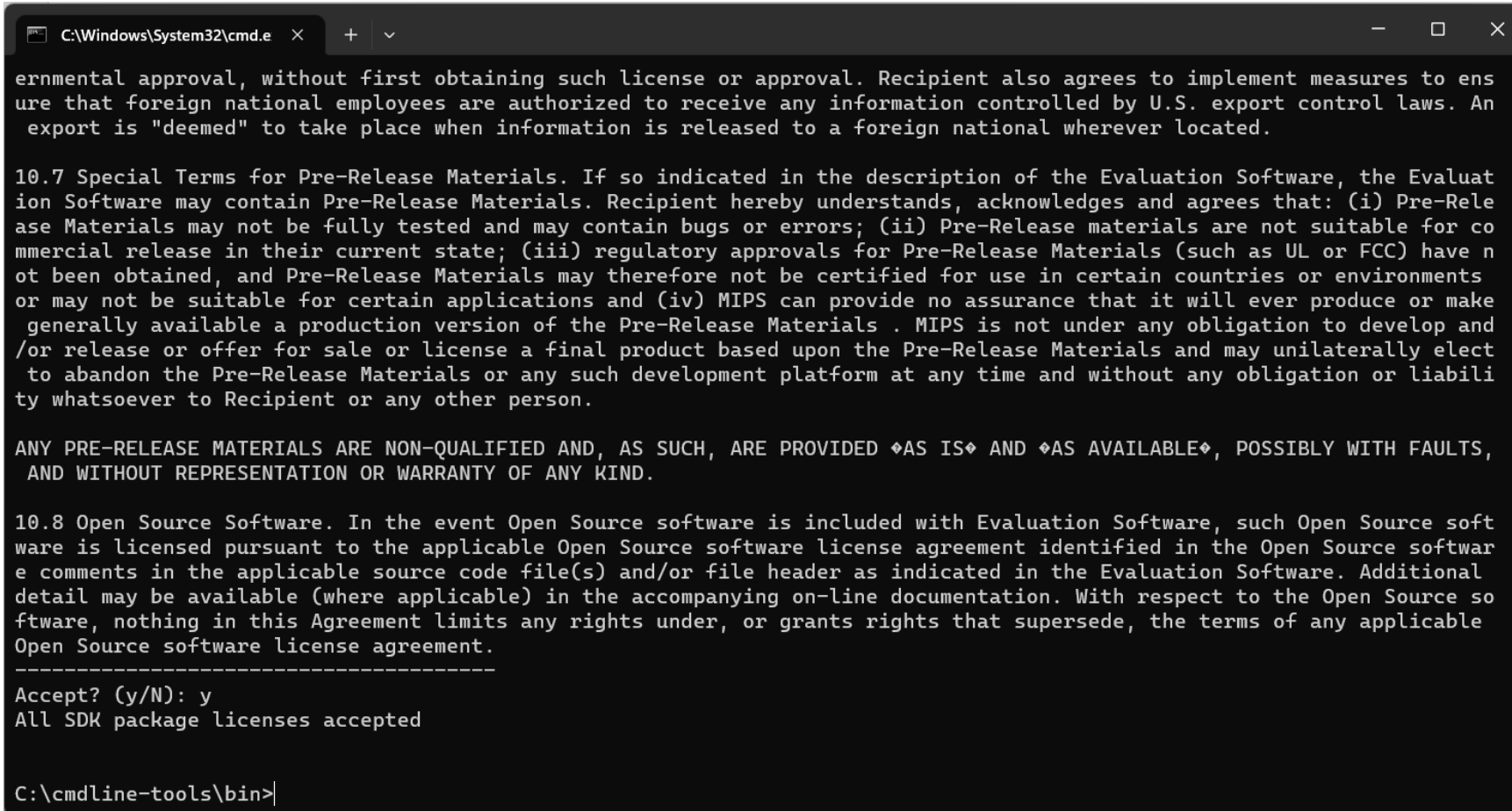
k. Install android licenses

- flutter doctor --android-licenses
- Total ada sekitar 6 s.d. 7 persetujuan (y)

A screenshot of a Windows Command Prompt window. The title bar shows 'C:\Windows\System32\cmd.e' with standard window controls. The command prompt shows the command 'C:\cmdline-tools\bin>flutter doctor --android-licenses' being executed. The output is: '[=====] 100% Computing updates...' followed by '6 of 7 SDK package licenses not accepted.' and a prompt 'Review licenses that have not been accepted (y/N)? |' with a cursor at the end.

```
C:\cmdline-tools\bin>flutter doctor --android-licenses
[=====] 100% Computing updates...
6 of 7 SDK package licenses not accepted.
Review licenses that have not been accepted (y/N)? |
```

7. Instalasi Android SDK



```
C:\Windows\System32\cmd.e x + v

ernmental approval, without first obtaining such license or approval. Recipient also agrees to implement measures to ens
ure that foreign national employees are authorized to receive any information controlled by U.S. export control laws. An
export is "deemed" to take place when information is released to a foreign national wherever located.

10.7 Special Terms for Pre-Release Materials. If so indicated in the description of the Evaluation Software, the Evaluat
ion Software may contain Pre-Release Materials. Recipient hereby understands, acknowledges and agrees that: (i) Pre-Rele
ase Materials may not be fully tested and may contain bugs or errors; (ii) Pre-Release materials are not suitable for co
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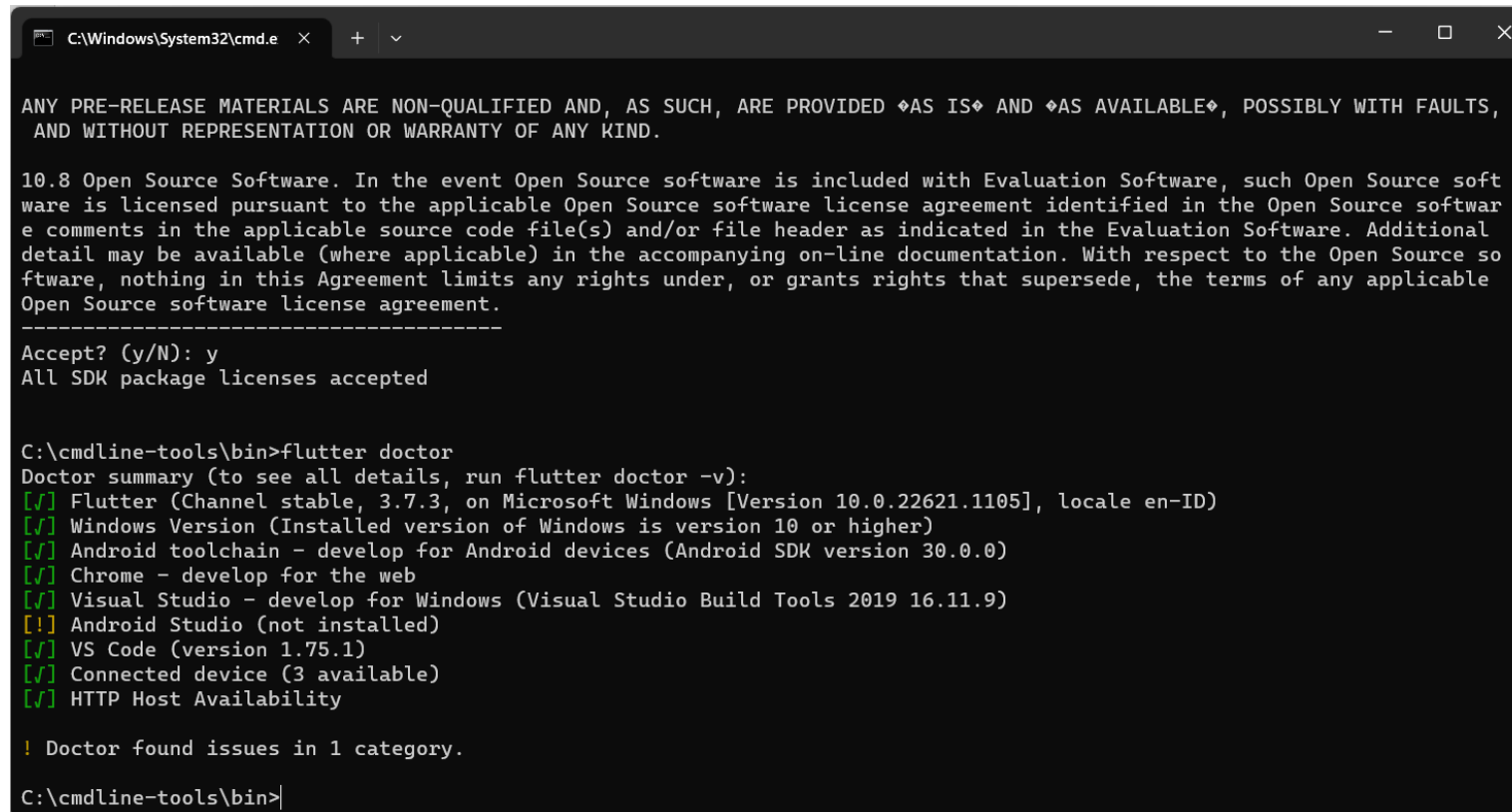
ANY PRE-RELEASE MATERIALS ARE NON-QUALIFIED AND, AS SUCH, ARE PROVIDED ♦AS IS♦ AND ♦AS AVAILABLE♦, POSSIBLY WITH FAULTS,
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10.8 Open Source Software. In the event Open Source software is included with Evaluation Software, such Open Source soft
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e comments in the applicable source code file(s) and/or file header as indicated in the Evaluation Software. Additional
detail may be available (where applicable) in the accompanying on-line documentation. With respect to the Open Source so
ftware, nothing in this Agreement limits any rights under, or grants rights that supersede, the terms of any applicable
Open Source software license agreement.
-----
Accept? (y/N): y
All SDK package licenses accepted

C:\cmdline-tools\bin>
```

7. Instalasi Android SDK

- flutter doctor



```
C:\Windows\System32\cmd.e x + v

ANY PRE-RELEASE MATERIALS ARE NON-QUALIFIED AND, AS SUCH, ARE PROVIDED ♦AS IS♦ AND ♦AS AVAILABLE♦, POSSIBLY WITH FAULTS,
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10.8 Open Source Software. In the event Open Source software is included with Evaluation Software, such Open Source soft
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e comments in the applicable source code file(s) and/or file header as indicated in the Evaluation Software. Additional
detail may be available (where applicable) in the accompanying on-line documentation. With respect to the Open Source so
ftware, nothing in this Agreement limits any rights under, or grants rights that supersede, the terms of any applicable
Open Source software license agreement.
-----
Accept? (y/N): y
All SDK package licenses accepted

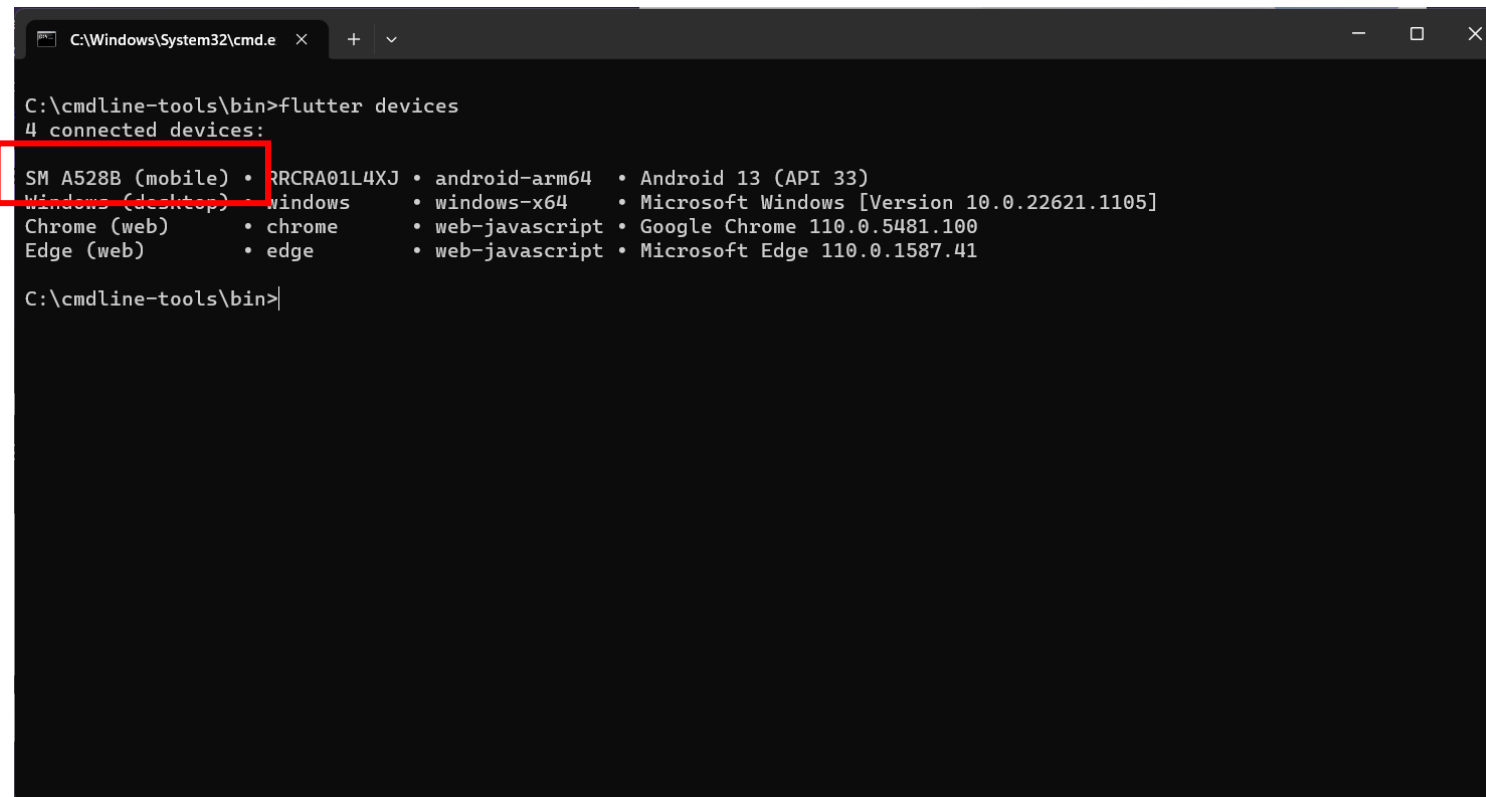
C:\cmdline-tools\bin>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.7.3, on Microsoft Windows [Version 10.0.22621.1105], locale en-ID)
[✓] Windows Version (Installed version of Windows is version 10 or higher)
[✓] Android toolchain - develop for Android devices (Android SDK version 30.0.0)
[✓] Chrome - develop for the web
[✓] Visual Studio - develop for Windows (Visual Studio Build Tools 2019 16.11.9)
[!] Android Studio (not installed)
[✓] VS Code (version 1.75.1)
[✓] Connected device (3 available)
[✓] HTTP Host Availability

! Doctor found issues in 1 category.

C:\cmdline-tools\bin>
```

7. Instalasi cmdline-tools

- Koneksikan smartphone ke computer menggunakan kabel data
- ketik perintah flutter devices



```
C:\Windows\System32\cmd.e  x  +  v

C:\cmdline-tools\bin>flutter devices
4 connected devices:
SM A528B (mobile) • RRCRA01L4XJ • android-arm64 • Android 13 (API 33)
Windows (desktop) • windows • windows-x64 • Microsoft Windows [Version 10.0.22621.1105]
Chrome (web) • chrome • web-javascript • Google Chrome 110.0.5481.100
Edge (web) • edge • web-javascript • Microsoft Edge 110.0.1587.41

C:\cmdline-tools\bin>
```

7. Instalasi cmdline-tools

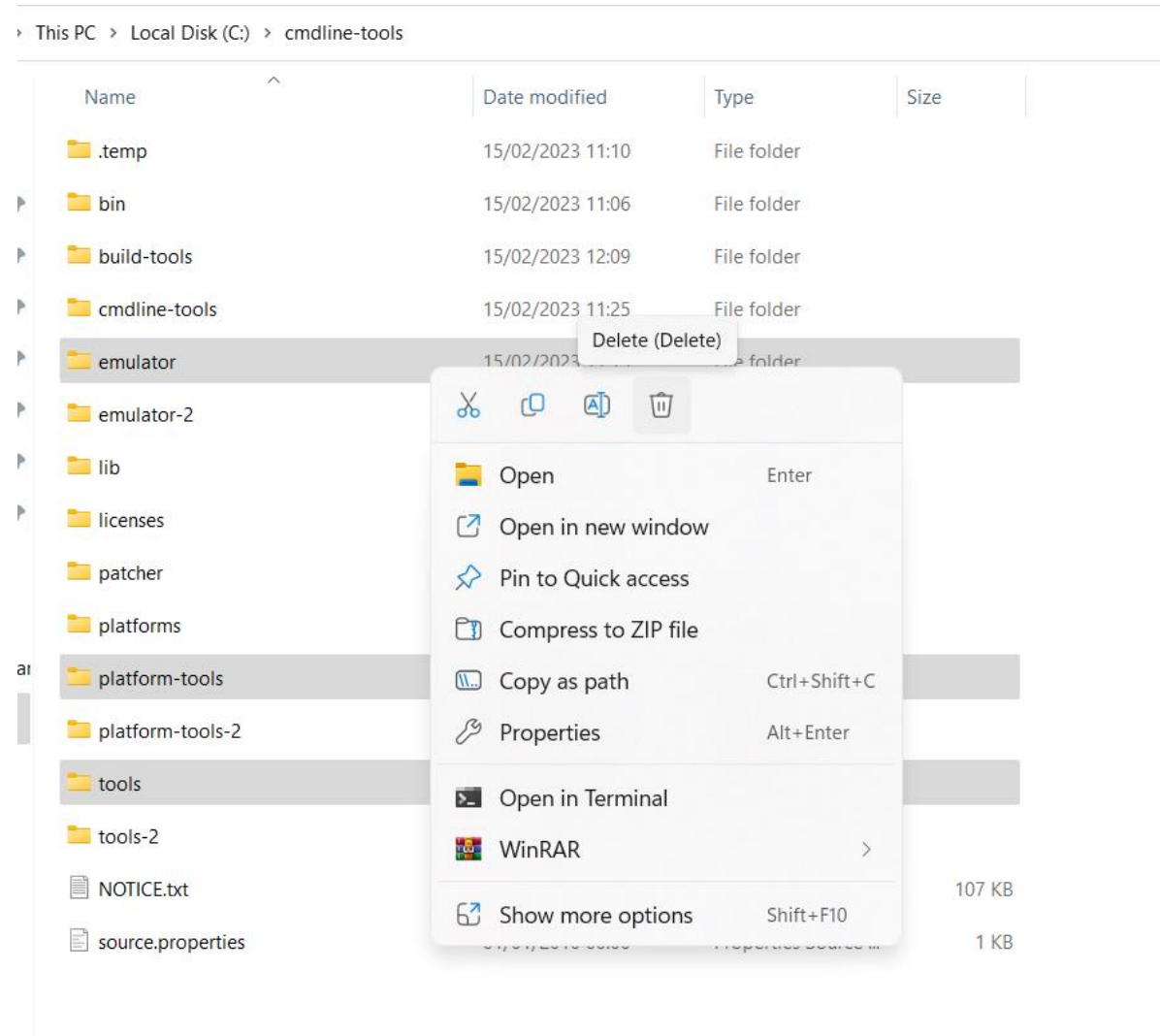
- Running project
- Untuk instalasi build-tools dan platform versi 30 akan muncul error
- (normal) 😊

PROBLEMS	OUTPUT	DEBUG CONSOLE	TERMINAL	COMMENTS
		<pre>License for package Android SDK Platform 33 accepted. Preparing "Install Android SDK Platform 33 (revision: 2)". "Install Android SDK Platform 33 (revision: 2)" ready. Installing Android SDK Platform 33 in C:\cmdline-tools\platforms\android-33 "Install Android SDK Platform 33 (revision: 2)" complete. "Install Android SDK Platform 33 (revision: 2)" finished. FAILURE: Build failed with an exception. * What went wrong: Could not determine the dependencies of task ':app:compileDebugJavaWithJavac'. > Failed to find target with hash string 'android-33' in: C:\cmdline-tools * Try: > Run with --stacktrace option to get the stack trace. > Run with --info or --debug option to get more log output. > Run with --scan to get full insights. * Get more help at https://help.gradle.org BUILD FAILED in 2m 42s Exception: Gradle task assembleDebug failed with exit code 1 Exited (sigterm)</pre>		

>

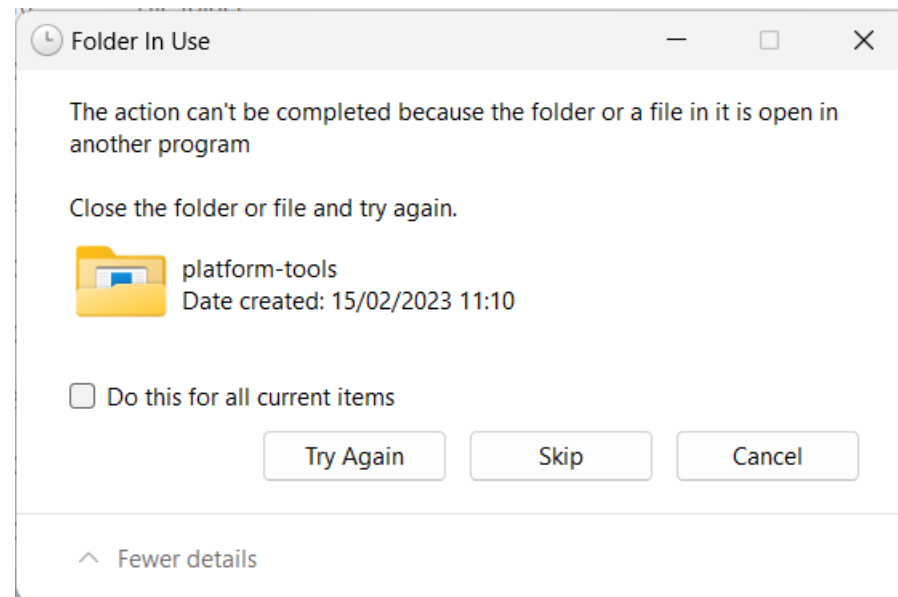
7. Instalasi cmdline-tools

- Buka folder cmdline-tools
- Hapus folder yang lama:
 - emulator
 - platform-tools
 - tools



7. Instalasi cmdline-tools

- Jika muncul dialog pesan, maka pastikan semua program terkait telah di-close



7. Instalasi cmdline-tools

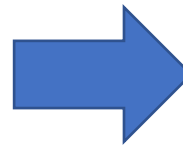
- Pastikan semua program terkait telah di-close
 - Terminal
 - VSCode
 - Eject smartphone
 - Task manager → Java TM Platform binary (end task)
 - Task manager → adb.exe (32 bit)

7. Instalasi cmdline-tools

- Rename semua folder yang ada angkanya
 - emulator-2 ➔ emulator
 - dst

> This PC > Local Disk (C:) > cmdline-tools >

Name	Date modified	Type	Size
.temp	15/02/2023 11:10	File folder	
bin	15/02/2023 11:06	File folder	
build-tools	15/02/2023 12:09	File folder	
emulator-2	15/02/2023 12:08	File folder	
lib	15/02/2023 11:06	File folder	
licenses	15/02/2023 11:29	File folder	
patcher	15/02/2023 12:06	File folder	
platforms	15/02/2023 12:09	File folder	
platform-tools-2	15/02/2023 12:06	File folder	
tools-2	15/02/2023 12:09	File folder	
NOTICE.txt	01/01/2010 00:00	Text Document	107 KB
source.properties	01/01/2010 00:00	Properties Source ...	1 KB

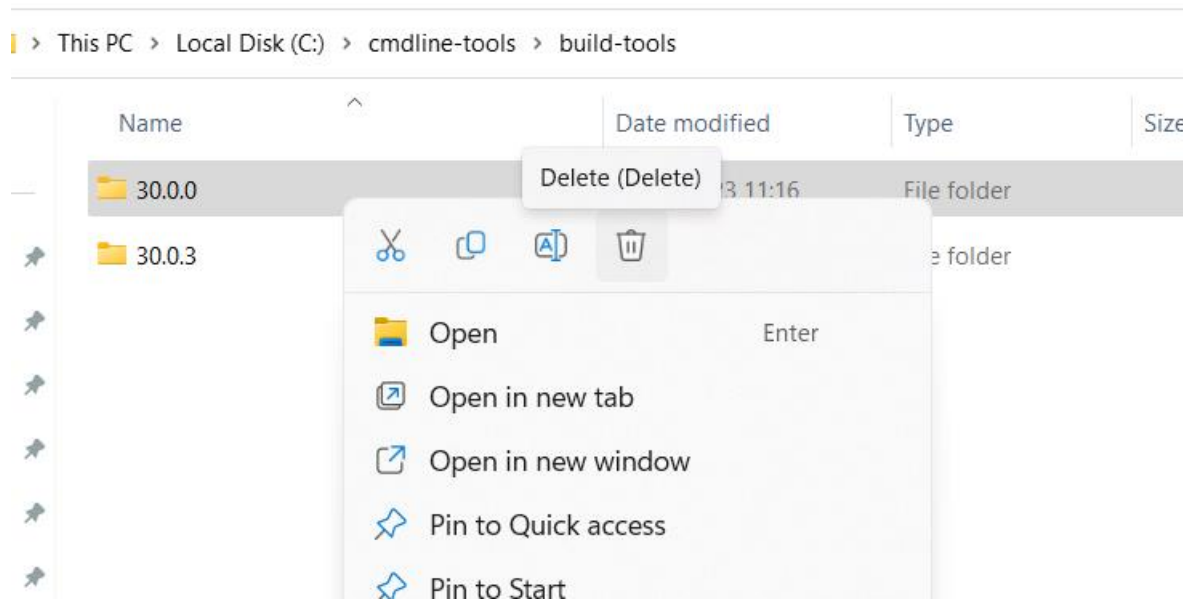


> This PC > Local Disk (C:) > cmdline-tools >

Name	Date modified	Type	Size
.temp	15/02/2023 11:10	File folder	
bin	15/02/2023 11:06	File folder	
build-tools	15/02/2023 12:09	File folder	
emulator	15/02/2023 12:08	File folder	
lib	15/02/2023 11:06	File folder	
licenses	15/02/2023 11:29	File folder	
patcher	15/02/2023 12:06	File folder	
platforms	15/02/2023 12:09	File folder	
platform-tools	15/02/2023 12:06	File folder	
tools	15/02/2023 12:09	File folder	
NOTICE.txt	01/01/2010 00:00	Text Document	107 KB
source.properties	01/01/2010 00:00	Properties Source ...	1 KB

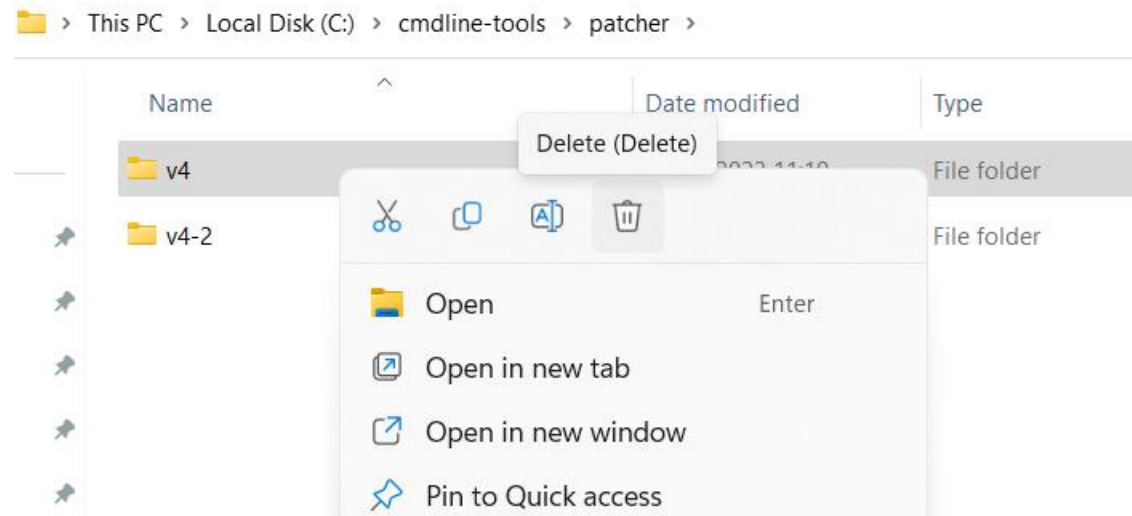
7. Instalasi cmdline-tools

- Pada folder build-tools, hapus versi yang lama (menghemat storage)



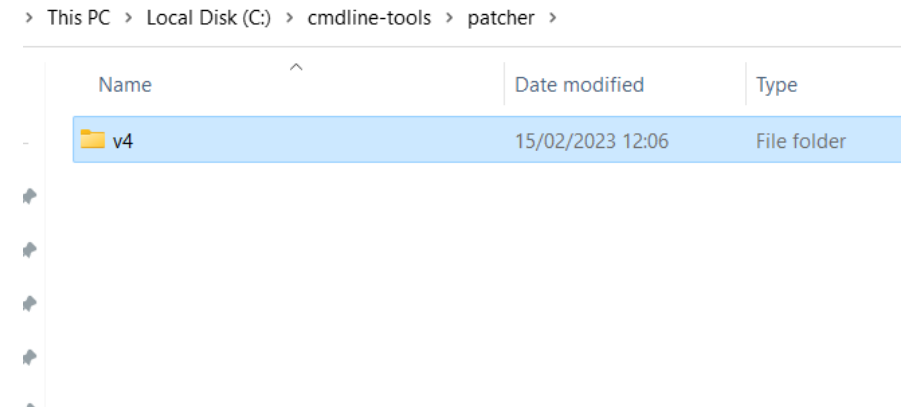
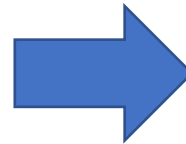
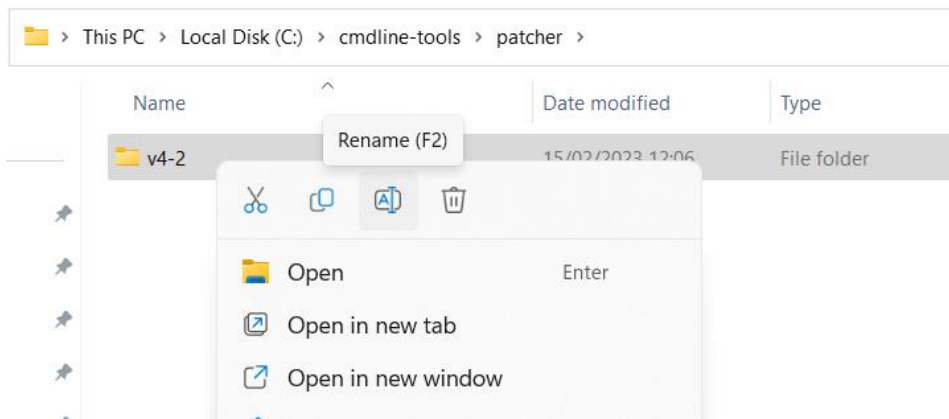
7. Instalasi cmdline-tools

- Hapus yang lama (menghemat storage)



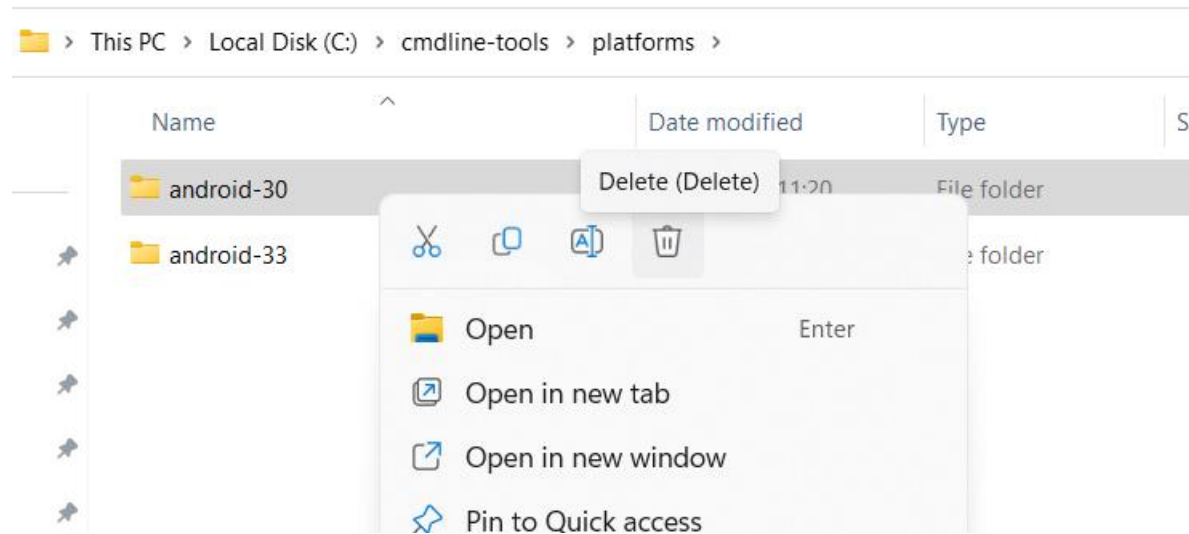
7. Instalasi cmdline-tools

- Rename yang baru



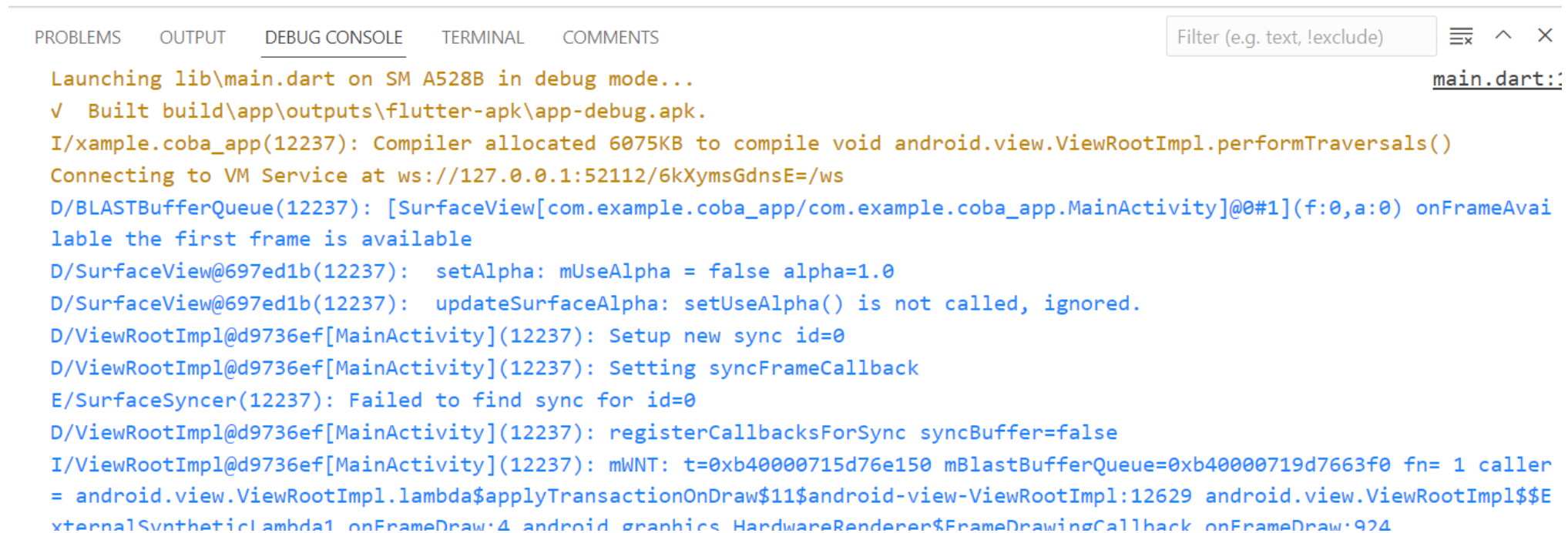
7. Instalasi cmdline-tools

- Hapus yang lama



7. Instalasi cmdline-tools

- Running kembali project flutter
- Sampai tahap ini, Aplikasi flutter android sudah bisa running



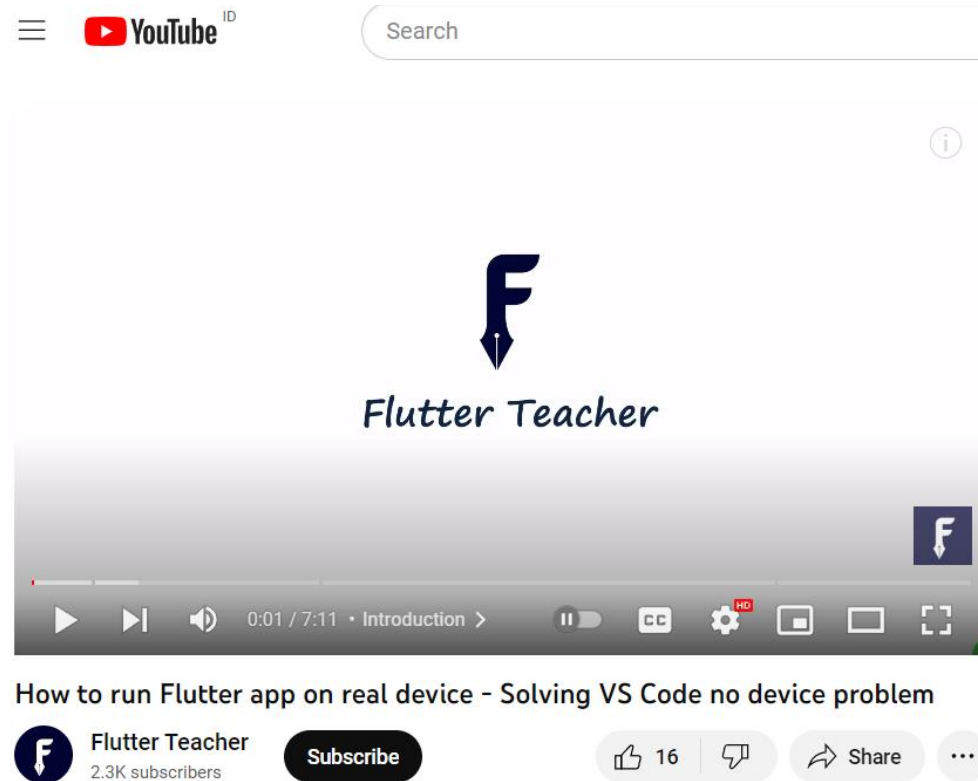
The screenshot shows the Android Studio interface with the 'DEBUG CONSOLE' tab selected. The output shows the process of launching the Flutter application in debug mode on a physical device (SM A528B). The log includes messages about building the APK, connecting to the VM service, and the initial rendering of the first frame. The application is running successfully.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL COMMENTS Filter (e.g. text, lexclude) main.dart:
Launching lib\main.dart on SM A528B in debug mode...
✓ Built build\app\outputs\flutter-apk\app-debug.apk.
I/xample.coba_app(12237): Compiler allocated 6075KB to compile void android.view.ViewRootImpl.performTraversals()
Connecting to VM Service at ws://127.0.0.1:52112/6kXymsGdnsE=/ws
D/BLASTBufferQueue(12237): [SurfaceView[com.example.coba_app/com.example.coba_app.MainActivity]@0#1](f:0,a:0) onFrameAvai
lable the first frame is available
D/SurfaceView@697ed1b(12237): setAlpha: mUseAlpha = false alpha=1.0
D/SurfaceView@697ed1b(12237): updateSurfaceAlpha: setUseAlpha() is not called, ignored.
D/ViewRootImpl@d9736ef[MainActivity](12237): Setup new sync id=0
D/ViewRootImpl@d9736ef[MainActivity](12237): Setting syncFrameCallback
E/SurfaceSyncer(12237): Failed to find sync for id=0
D/ViewRootImpl@d9736ef[MainActivity](12237): registerCallbacksForSync syncBuffer=false
I/ViewRootImpl@d9736ef[MainActivity](12237): mWNT: t=0xb40000715d76e150 mBlastBufferQueue=0xb40000719d7663f0 fn= 1 caller
= android.view.ViewRootImpl.lambda$applyTransactionOnDraw$11$android-view-ViewRootImpl:12629 android.view.ViewRootImpl$$E
xternalSyntheticLambda1 onFrameDraw:4 android.graphics HardwareRenderer$FrameDrawingCallback onFrameDraw:924
```

9. Instalasi physical device (wired)

9. Instalasi physical device (wired)

- <https://www.youtube.com/watch?v=yxif9Tj8fDE>



10. Instalasi physical device (wireless)

10. Instalasi physical device (wireless)

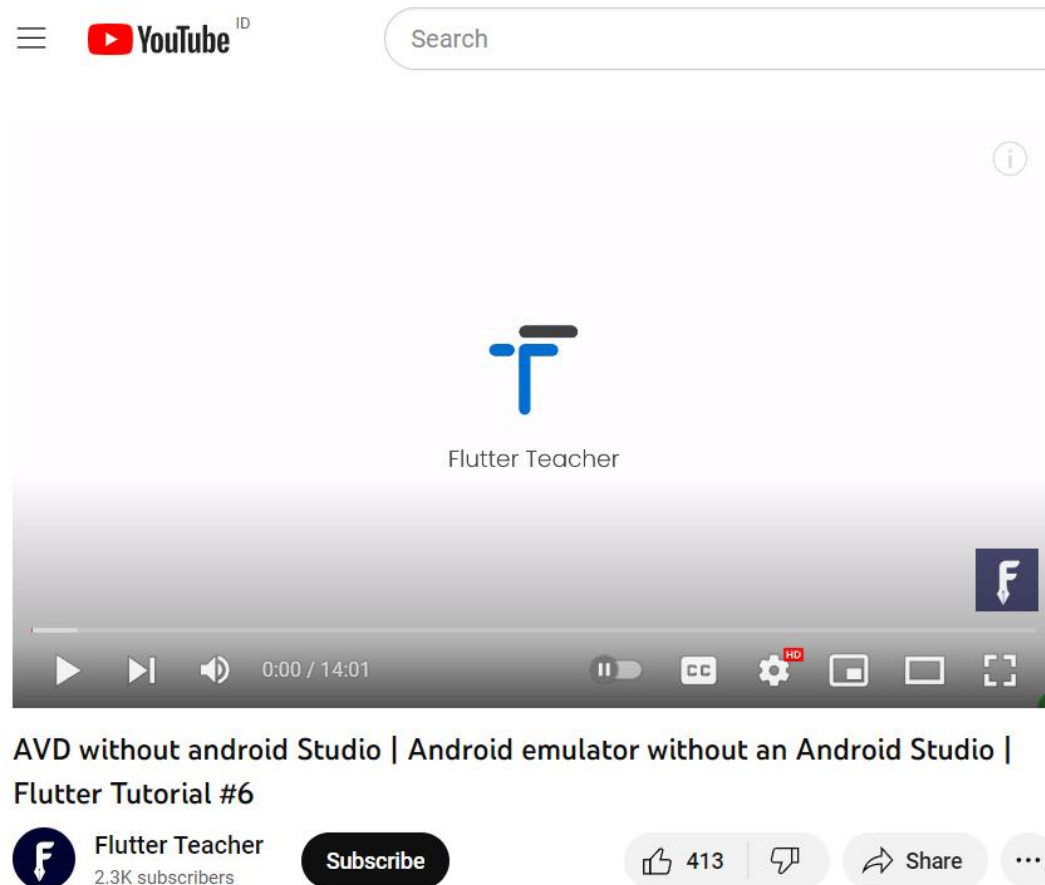
- <https://www.youtube.com/watch?v=p2bsfBA6lwg>



11. Instalasi virtual device (emulator)

11. Instalasi virtual device (emulator)

- <https://www.youtube.com/watch?v=591Sme4jxDc>



Selesai